multiomicsGEP — Project Status

Supervised Bayesian Matrix Factorization for Joint Genomics + Survival Modelling

Andrew Walther

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0.1 Correction: CPTAC is RNA-seq (2026-10-16)

The CPTAC training cohort (n = 129) is RNA-seq, not proteomics as earlier versions of this document stated. It is supplied on the log2 scale, whereas TCGA_PAAD is supplied untransformed; preprocessing was already correct. The pooled-training collapse without per-cohort standardization stands as an empirical finding, but it reflects a between-cohort technical difference between two RNA-seq datasets, not an assay difference. Details and the list of affected records are in the 2026-10-16 correction entry of `DECISIONS.md`.

0.2 Project Overview

This project implements a **Supervised Bayesian Matrix Factorization** model that jointly decomposes high-dimensional genomics data (gene expression, DNA methylation, etc.) and models patient survival outcomes. The core idea is to find a shared latent space that simultaneously explains genomic variation *and* predicts clinical outcomes.

The model:

- **Genomics:** Y = L F' + E, where L (n x K) are patient loadings, F (p x K) are biological factor weights, and E is Gaussian noise with feature-specific precision tau_j.

- **Survival:** Cox proportional hazards h(t_i) = h_0(t_i) exp(sum_k l_ik * beta_k), linking the same latent loadings L to time-to-event data via survival coefficients beta.

Inference: Coordinate Ascent Variational Inference (CAVI), where each parameter update reduces to an Empirical Bayes Normal Means (EBNM) problem with point-normal priors.

0.3 Matched expression and methylation: derivation for review (2026-09-10)

The multimodal YFB mathematical proposal and its Quarto source adapt the existing YFB parameterization to one training cohort with matched expression and methylation and one independent matched validation cohort. The model shares subject scores and survival coefficients, while fitting separate loading priors and feature-wise residual precisions by modality.

The concise nine-page proposal derives the EBMF plus survival objective and L/F/β/Tau updates for linear joint YFB using the diagonal Cox working approximation. Separate point-exponential loading priors are estimated per modality and factor. The implementation specification maps numbered equations to proposed R updates and tests. The original extensive derivation remains in the same folder as technical notes. All 35 existing algebra checks pass, including loading interactions and posterior second moments.

Following approval, an isolated runnable multimodal fitter, frozen predictor, matched-block simulation generator, and checkpointed K-selection runner were implemented under `code/` and tested through `tests/run_tests.R`. The established single-modality production model is unchanged. The multimodal fitter uses factor-scale canonicalization and a relative identifiable convergence criterion (maximum 500 sweeps); its K=5,…,15 replicated held-out selection study is the remaining prerequisite to choosing a multimodal default. The September 18 progress-book chapter records the implementation and preliminary screening.

2026-10-01/02: choosing K and the first real-data application.

Why the number of active factors tracked K_{init}:

- Unregularized per-feature precisions let a spare factor fit a single feature exactly, so its precision was unbounded (up to 10^{16} in simulation).
- The model had no intercept, so spare nonnegative factors absorbed feature means.
- In the September simulations each true factor explained under 1% of the variance, so the design could not test this.

The fix is a parsimony framework: a per-feature intercept, point-Laplace loadings, `ebnm` marginal-likelihood prior fits, and ELBO-based factor pruning using the Cox partial log-likelihood. In single-seed simulations at moderate noise it recovers the true $K = 3$ from both $K_{\text{init}} = 5$ and 12, with the prognostic programs survival-active and of the correct sign. A low-variance prognostic program is recovered only partially.

Matched TCGA □ ICGC expression + methylation data are loaded and tested (`code/load_multiomics_data.R`). The replicated simulation sweep ($K_{\text{init}} = 3, \ldots, 10, 15$) and the real-data fits are recorded in the 2026-10-02 progress-book chapter. Decisions are in `DECISIONS.md` (2026-10-01, 2026-10-02).

Single-modality reporting was checked in the same period. The external C-indices are unchanged under frozen, training-fixed risk orientations. The pooled improvement over the two-step baseline is $\Delta C = 0.042$ (95% CI 0.013 to 0.071).

0.4 Advisor-facing project writeup (2026-10-14)

The unified methods, results, and biology writeup is being organized in `paper/writeup/`. Its planning page describes an article-style technical report with a 15–20-page main text and separate single-modality and multimodal derivation appendices, rendered together to HTML and PDF. The source inventory maps 54 existing sources to planned sections and distinguishes checked numerical selections from material still requiring reconciliation during drafting. The implementation plan records the verification requirements and remaining tasks.

Claude Code's planning review has been checked against the cited sources; the review dispositions and revised plan specify characterization methods, completed versus proposed multimodal biology, a six-claim evidence-gap table, and source markers for the later supervised-F revision. The drafting handoff assigns Claude Tasks 2, 3 and 5 on `writeup/draft`; Codex performs the independent complete-draft review afterward.

Status (2026-10-14): interim writeup approved and merged to `main`. It describes the evaluated configuration (α_F = 0); REVISIT markers flag the passages the method work will change, and Prompt 7 of the 10/2 plan revises it afterwards. The main text is 22 pages, accepted for the interim version. Codex reviewed the initial draft (`paper/writeup/_process/review.md`); every finding was checked against code and saved outputs and applied, and dispositions are recorded in `paper/writeup/_process/review_dispositions.md`. The report is now organized as a manuscript (Methods, Evaluation, Results, Biological findings, Discussion with implications, limitations and continuing work), without source IDs, repository references, dated process language, an evidence-gap table or a bibliography (DeSurv is attributed as Young et al., submitted). Appendices A and B derive every update from the likelihood, priors and variational family to the EBNM inputs. Scientific corrections from the review: the Cox working curvature is a Poisson-style diagonal replacement, not the Hessian diagonal; the single-modality stopping statistic is a monitor (its activity KL term counts the Gaussian normalizer once instead of n times); predictions use the norms stored at the start of the final sweep; the multimodal removal score uses KL penalties cached against earlier priors; the concatenated C is 0.6285 versus 0.6026. The two implementation defects (activity-KL normalizer, cached-prior KL) are documented, not fixed, and belong to the method work. The established model's matched-donor C = 0.685 is omitted (no saved source). All figures were redrawn from saved outputs as vector PDF and 300-dpi PNG. Codex then verified the revision (`paper/writeup/_process/verification.md`); its findings V1–V13 were checked and applied (dispositions in `paper/writeup/_process/review_dispositions.md`), including the zero-column normalized target, the protected multimodal precision floor, completed derivation steps, local caption definitions and unbroken algorithms and tables. Remaining issue: the PDF main text is 22 pages (down from 27) against a 15–20-page target, after a final editorial check (`paper/writeup/_process/final_editorial_check.md`) moved both algorithms and derivation-level detail into the appendices. No model-fitting experiment or algorithm change is part of this writing task. The second verification (`paper/writeup/_process/verification_round2.md`) confirmed the scientific corrections, derivations, captions and pagination. Its two minor corrections are applied: the first sweep-normalizer formulas now include the evaluated numerical ridge, and the disposition-file

pointer above is corrected. Figure regeneration still requires local held-out expression; further length cuts and final approval remain Andrew's decision. Existing prelim and historical derivation sources are preserved.

0.5 Repository Map

```
multiomicsGEP/
|-- CLAUDE.md                          <- Claude Code entry point (thin; defers here)
|-- PROJECT_STATUS.qmd/.pdf           <- This file (project documentation)
|-- README.md                          <- GitHub front page
|-- code/
|   |-- SupervisedMF_Context.md      <- AI/collaborator quick-reference for V2 code
|   |-- legacy/Supervised_Bayesian_MF.R <- V1: Original CAVI implementation (archived)
|   |-- Supervised_Bayesian_MF_V2.R  <- V2: Corrected implementation (all fixes) -- USE
|   |-- update_beta.R                <- Modular beta update functions
|   |-- update_L.R                   <- Modular q(L) update functions (vector EBNM, dua
source)
|   |-- update_F.R                   <- Modular q(F) update functions (pure genomics, t
|   |-- update_tau.R                 <- Modular q(tau) update functions (closed-
form MLE)
|   `-- multiomicsGEP_code.Rmd       <- Earlier exploratory R Markdown notebook
|-- derivations/
|   |-- qB/
|   |   |-- qBeta_update_derivation.tex  <- Full step-by-step q(beta) derivation (11 pa
|   |   `-- qBeta_update_derivation.pdf
|   |-- qL/
|   |   |-- qL_update_derivation.tex     <- Vector EBNM, dual-source (genomics+survival
|   |   `-- qL_update_derivation.pdf
|   |-- qF/
|   |   |-- qF_update_derivation.tex     <- tau cancellation property, pure genomics
|   |   `-- qF_update_derivation.pdf
|   |-- qTau/
|   |   |-- qTau_update_derivation.tex   <- Closed-form MLE, variance correction
|   |   `-- qTau_update_derivation.pdf
|   |-- EBMF/
|   |   `-- EBMF_Derivations*.pdf        <- Empirical Bayes MF theory notes
|   |-- MF_UpdateDerivations/
|   |   |-- MF_Derivations_UpdateAlgo_2_12_26.pdf  <- Reviewed draft -- HAS ERRORS (R1-
R8)
|   |   |-- MF_Derivations_UpdateAlgo_REVISED.tex/.pdf  <- Corrected (21 pages)
|   |   `-- MF_V2_Companion.tex/.pdf    <- Math <-> code mapping (17 pages)
|   `-- SurvivalMF/
|       `-- *.pdf                        <- Survival + MF background notes
|-- results/
|   |-- run_simulation.R             <- Reproduce V2 simulation (8 figs, 11 CSVs)
|   |-- run_modular_simulation.R     <- Reproduce modular simulation (standalone)
|   |-- simulation_report.qmd/.pdf  <- V2 simulation report (Quarto, ggplot2)
```

```
|   |-- modular_sim_report.qmd/.pdf <- Modular simulation report (Quarto, per-
fig commentary)
|   |-- simulation_console_output.txt
|   |-- figures/
|   |   |-- full_sim/               <- 8 figure pairs from V2 simulation
|   |   `-- modular_sim/            <- 8 figure pairs from modular simulation
|   `-- tables/
|       |-- full_sim/               <- 11 CSV tables from V2 simulation
|       `-- modular_sim/            <- 11 CSV tables from modular simulation
|-- tests/
|   |-- test_helpers.R              <- Lightweight assertion framework
|   |-- test_update_beta.R          <- 24 tests for update_beta.R (9 groups)
|   |-- test_update_L.R             <- 28 tests for update_L.R (9 groups)
|   |-- test_update_F.R             <- 26 tests for update_F.R (9 groups)
|   |-- test_update_tau.R           <- 27 tests for update_tau.R (9 groups)
|   `-- run_tests.R                 <- Master runner: 105/105 tests passing
|-- demos/
|   |-- demo_update_beta.R          <- 5 demos for update_beta.R
|   |-- demo_update_L.R             <- 5 demos for update_L.R
|   |-- demo_update_F.R             <- 5 demos for update_F.R
|   `-- demo_update_tau.R           <- 5 demos for update_tau.R
|-- docs/
|   |-- Makefile                    <- `make all` renders .qmd -> .pdf + .html via Qua
|   |-- update_beta.qmd/.pdf/.html  <- Companion doc for update_beta.R
|   |-- update_L.qmd/.pdf/.html     <- Companion doc for update_L.R
|   |-- update_F.qmd/.pdf/.html     <- Companion doc for update_F.R
|   `-- update_tau.qmd/.pdf/.html   <- Companion doc for update_tau.R
`-- paper/
    `-- multiomicsGEP_manuscript.qmd  <- Manuscript draft (Quarto)
```

0.6 Mathematical Framework

0.6.1 Model Components

Symbol	Dimension	Meaning
Y	n x p	Observed genomics data matrix
L	n x K	Patient loading matrix (shared latent space)
F	p x K	Biological factor weight matrix
beta	K x 1	Survival coefficient vector
tau_j	p-vector	Feature-specific noise precision
(t_i, delta_i)	n-vectors	Survival time and event indicator

0.6.2 Inference Strategy

1. **Mean-field factorisation:** q(L, F, beta) = q(L) q(F) q(beta)
2. **CAVI loop:** For each factor k = 1, …, K:
   - Update q(l_k) via EBNM(x = B_L/A_L, s = 1/sqrt(A_L))
   - Update q(f_k) via EBNM(x = B_F/A_F, s = 1/sqrt(A_F))
   - Update q(beta_k) via EBNM(x = B_beta/A_beta, s = 1/sqrt(A_beta))
3. **Update tau_j** from expected squared residuals
4. **Cox Taylor expansion** linearises the non-conjugate survival likelihood into a weighted Gaussian form with working response z_i and weight W_ii

0.6.3 Key Mathematical Concepts

- **Posterior means vs second moments:** EL = E_q[l], EL2 = E_q[l^2] = Var_q(l) + mean^2. This distinction is critical for the variance inflation term in the tau update and the error-in-variables correction in the beta update.
- **EBNM sub-problems:** Each coordinate update reduces to: given pseudo-observations x with noise s, estimate a prior g and posterior q.
- **Error-in-variables:** Using E[l^2] (not l-bar^2) in the beta precision prevents overfitting to uncertain loadings.

0.7 Derivation Review Summary

A thorough review of `MF_Derivations_UpdateAlgo_2_12_26.pdf` identified **8 errors**, all corrected in `MF_Derivations_UpdateAlgo_REVISED.tex`.

> ⚠ Derivation errors in 2_12_26.pdf (R1–R8)
>
> The February 12 draft contains 8 errors listed below. Use **REVISED.pdf** for all reference.

ID	Location	Error	Correction
R1	Sec. 1A	Wrong dimension subscripts on Y, L, F, E	Fixed: Y_(nxp) = L_(nxK) F'*(Kxp)* + *E*(nxp)
R2	Eq. 22	z^{-k}*i defined with f*{jk'} instead of beta_{k'}	Fixed: uses beta_{k'} (survival coefficient)
R3	Eqs. 32-35	f^2_{jk}, beta^2_k where E_q second moments needed	Added explicit E_q notation

ID	Location	Error	Correction
R4	Eqs. 47-48	EBNM inputs subscripted i (sample)	Fixed to j (feature) for F update
R5	Eq. 62	Missing exponent: beta_k should be beta^2_k	Added squared exponent
R6	Sec. 6	beta update as generic WLS; no EBNM form	Derived full A_k, B_k for EBNM
R7	Eq. 83	Sign error: log tau + tau * R-bar^2	Fixed to log tau - tau * R-bar^2
R8	Eq. 85	l^2_{jk} wrong index (j should be i)	Fixed to l^2_{ik}

0.7.1 Document Relationships

- **2_12_26.pdf** — reviewed document; contains errors R1–R8
- **REVISED.tex/.pdf** — corrected version with all 8 fixes + errata table (21 pages)
- **MF_V2_Companion.tex/.pdf** — comprehensive companion mapping math to V2 code (17 pages)

0.8 Code Versions

0.8.1 V1: `code/legacy/Supervised_Bayesian_MF.R`

The original implementation. Functionally correct but with several known issues:

- Orthogonalisation always ON; resets EL2/EF2 to squared means (drops posterior variance every 10 iterations)
- z_no_k computed from start-of-iteration eta only (not true Gauss-Seidel)
- No numerical floors on EBNM precision inputs
- Convergence checks only delta-L (not delta-beta)
- Only RMSE tracked (no ELBO monitoring)

0.8.2 V2: `Supervised_Bayesian_MF_V2.R`

> Recommended implementation
>
> Use `code/Supervised_Bayesian_MF_V2.R` for all runs. V1 is retained for reference only.

The corrected implementation with 6 algorithmic improvements:

ID	Change	Description
A1	Gauss-Seidel CAVI	z_no_k recomputed from current EL, EBeta inside k-loop
A2	Orthogonalisation flag	Behind `orthogonalize=FALSE` (default OFF)
A3	Precision floors	`pmax(..., 1e-10)` on A_L, A_F, A_Beta
A4	Dual convergence	Both mean
A5	ELBO proxy	Genomics log-likelihood tracked per iteration
A6	Taylor refresh flag	`refresh_taylor=FALSE` for optional per-k recomputation

> **Convergence in the canonical modular loops (2026-06-16 clarification).** The A4 "dual delta-L / delta-beta" rule describes the V2 monolithic reference. The canonical modular CAVI loops used for all current results — `code/fit_modular.R` (LB) and `code/fit_cox_on_yf.R` (YFB) — instead declare convergence on the **relative full-ELBO change < tol (default 1e-5)**, after a 5-iteration burn-in; mean|delta-L| and mean|delta-beta| are computed and logged as diagnostics but are not the stopping rule. The ELBO is both the optimization objective and the convergence monitor; the number of programs K is selected separately by cross-validated C-index (1-SE rule). See DECISIONS.md 2026-06-16.

Function signature:

```
fit_supervised_mf(Y, time, status, K=5, max_iter=100, tol=1e-5,
                  orthogonalize=FALSE, refresh_taylor=FALSE, verbose=TRUE)
# Returns: list(L, F, Beta, Beta2, Tau, history)
```

0.9 How to Run

0.9.1 Prerequisites

```
install.packages(c("survival", "ebnm"))
```

0.9.2 Running the Simulations

V2 (monolithic):

```
source("code/Supervised_Bayesian_MF_V2.R")
```

Modular (standalone, recommended for understanding the algorithm):

```
Rscript results/run_modular_simulation.R
```

Both use the same parameters (n=250, p=1000, K=5, seed=42) and produce equivalent results. The modular script sources only the four standalone update modules; it does not depend on V2.R.

0.9.3 Expected Output

- **RMSE** converges near 1.0 (the true noise SD)
- **ELBO proxy** is non-decreasing
- **Estimated beta** recovers sign pattern (+, -, +, -, 0)
- **Supervised C-index** ~0.86 on held-out tertiles
- **Censoring rate** ~30-40%

0.9.4 Compiling LaTeX Documents

```
cd derivations/MF_UpdateDerivations/
pdflatex MF_Derivations_UpdateAlgo_REVISED.tex
pdflatex MF_V2_Companion.tex
pdflatex MF_V2_Companion.tex   # second pass for cross-references

cd derivations/qB/
pdflatex qBeta_update_derivation.tex  # run 3x for stable bookmarks
```

0.9.5 Running the Test Suite

> Test suite status: 105/105 passing
>
> Run after any change to a modular update script: `Rscript tests/run_tests.R`

```
Rscript tests/run_tests.R
# Expected: 105/105 tests passed (24 beta + 28 L + 26 F + 27 tau)
```

0.9.6 Running the Modular Update Demos

```
Rscript demos/demo_update_beta.R   # beta update: 5 demos
Rscript demos/demo_update_L.R      # q(L) update: 5 demos
Rscript demos/demo_update_F.R      # q(F) update: 5 demos -- see tau cancellation
Rscript demos/demo_update_tau.R    # q(tau) update: 5 demos -- see variance correction
```

0.10 Current Status & Next Steps

0.10.1 Completed

- ☒ Derivation review: 8 errors (R1-R8) identified and corrected
- ☒ REVISED.tex/PDF: Corrected derivation document with errata table (21 pages)
- ☒ REVISED.tex expanded: Full step-by-step algebra added throughout (Taylor, L/F/beta/tau updates) with new `derivbox` gray-box environment for intermediate steps
- ☒ V2.R: Corrected R implementation with all 6 algorithmic fixes (A1-A6)
- ☒ V2_Companion.tex/PDF: Comprehensive math-to-code companion document (17 pages)
- ☒ PROJECT_STATUS + SupervisedMF_Context.md: Project documentation
- ☒ Simulation validation: V2 runs end-to-end on synthetic data (n=250, p=1000, K=5)
- ☒ results/: Full simulation outputs – 8 figures, 11 CSV tables, console log
- ☒ derivations/qB/qBeta_update_derivation.tex/.pdf: Self-contained q(beta) derivation (11 pages)
- ☒ code/update_beta.R: Modular beta update (`compute_z_no_k`, `update_beta_k`, `update_beta_all`)
- ☒ tests/: Test infrastructure + 24 tests for beta update (9 groups, all passing)
- ☒ demos/demo_update_beta.R: 5 interactive demos for beta update
- ☒ code/update_L.R: Modular q(L) update – vector EBNM, dual-source
- ☒ code/update_F.R: Modular q(F) update – pure genomics, tau cancellation
- ☒ code/update_tau.R: Modular q(tau) update – closed-form MLE
- ☒ tests/test_update_L.R: 28 tests (9 groups, all passing)
- ☒ tests/test_update_F.R: 26 tests (9 groups, tau cancellation verified)
- ☒ tests/test_update_tau.R: 27 tests (9 groups, variance correction verified)
- ☒ tests/run_tests.R: Updated master runner – 105/105 tests passing
- ☒ demos/demo_update_L.R: 5 demos (anatomy, signal recovery, genomics vs combined, error-in-variables, K=5)
- ☒ demos/demo_update_F.R: 5 demos (tau cancellation, sparse recovery, differential shrinkage, sparsity, K=5)
- ☒ demos/demo_update_tau.R: 5 demos (anatomy, known noise, variance correction, heteroscedastic, ELBO proxy)
- ☒ derivations/qL/qL_update_derivation.tex/.pdf: Self-contained q(L) derivation
- ☒ derivations/qF/qF_update_derivation.tex/.pdf: Self-contained q(F) derivation
- ☒ derivations/qTau/qTau_update_derivation.tex/.pdf: Self-contained q(tau) derivation
- ☒ docs/: Companion documentation for all 4 modular update scripts (.md + .pdf + .html)

- ☒ roxygen2 @export/@family/@seealso tags added to all 11 functions
- ☒ Targeted inline comments added to all 4 R scripts
- ☒ results/ reorganized: figures/{full_sim,modular_sim}/ and tables/{full_sim,modular_sim}/
- ☒ results/run_modular_simulation.R: Standalone CAVI script sourcing only the 4 modular modules
- ☒ results/modular_sim_report.qmd/.pdf: Quarto simulation report for modular implementation
- ☒ results/simulation_report.qmd/.pdf: Quarto simulation report for V2 implementation
- ☒ CLAUDE.md created; PROJECT_STATUS converted to .qmd for PDF rendering

0.10.2 Potential Next Steps

> **Note**
>
> **List maintenance:** Mark items `[x]` when implemented and confirmed, or explicitly dropped. Do not remove them — checked items are a record of accomplishments. Only remove true duplicates or items added in error.

Near-term (next 1–2 sessions):

- ☐ **Hold-out prediction task:** Train on 80% of each cohort, evaluate C-index on 20% held-out. Requires `predict_supervised_mf()`: fix learned F and β, run a single EBNM pass on Y_test to recover L_test, compute L_test %*% EBeta as linear predictor. Highest-priority performance validation.
- ☐ **K selection:** K=5 is fixed across all cohorts; likely suboptimal for small-n cohorts (Dijk n=90, PACA_AU_seq n=52). Options, roughly by cost:
  1. *Fit large K and prune* – fit K=10, let EBNM zero uninformative factors; count surviving non-zero factors as effective K. Free at inference time.
  2. *Scree plot / PVE elbow* – plot per-factor explained variance from existing `factor_summary_table.csv`; identify natural elbow. Zero additional compute.
  3. *Held-out C-index CV* – k-fold over candidate K values; computationally intensive, best run on Longleaf HPC.
- ☐ **Lab meeting slide deck (April 9, 2026):** Convert PDAC report to a Quarto revealjs presentation covering: model motivation, CAVI overview, per-dataset results, cross-dataset summary, pooled fit, and future directions.

Medium-term:

- ☐ **Survival-aware feature selection:** Replace top-N-by-variance with a univariate Cox prefilter or survival-weighted variance metric, to bias the input feature set toward prognostically relevant genes before factorization.
- ☐ **Batch correction before pooling:** Apply ComBat or `limma::removeBatchEffect` before `pool_datasets()` to reduce dataset-of-origin effects in the pooled RNA-seq model.

- ☐ **Prior family exploration:** Test `prior_family = "point_laplace"` or `"normal_scale_mixture"` in EBNM calls; may recover denser or more graded factor structures on real data vs. the default point-normal.
- ☐ **Full ELBO tracking:** Currently only the genomics log-likelihood proxy is tracked. Including the survival Taylor term and all KL divergences enables proper model selection and tighter convergence monitoring.
- ☐ **Multiple random initializations:** SVD initialization is deterministic. Running from multiple random starts and selecting the best ELBO would reduce sensitivity to local optima, at the cost of K× compute.

Longer-term:

- ☐ **Multi-dataset shared factor space:** Shared F and β with per-cohort L and τ. Requires new update equations and extension of `fit_modular.R`.
- ☐ **Multi-modal integration:** Each modality (expression, methylation) gets its own F^(m) and τ^(m) sharing a common L. Requires new `update_F_<m>.R` and extended `update_L.R` accumulating signal across modalities.
- ☐ **Manuscript:** Complete `paper/multiomicsGEP_manuscript.qmd` once prediction results and K selection are in hand.

0.11 Session Log

0.11.1 Session 1 (March 4–5, 2026)

Derivation review and code correction session.

1. Reviewed `MF_Derivations_UpdateAlgo_2_12_26.pdf` against the code
2. Identified 8 derivation errors (R1–R8)
3. Created `MF_Derivations_UpdateAlgo_REVISED.tex` with all corrections + errata table
4. Analysed V1 code against corrected derivations; identified 6 improvements
5. Created `Supervised_Bayesian_MF_V2.R` with all algorithmic fixes (A1–A6)
6. Created `MF_V2_Companion.tex` (math to code companion, 17 pages)
7. Created `PROJECT_STATUS.md` and `SupervisedMF_Context.md`

Key decisions:

- Orthogonalisation defaults to OFF in V2 (Point-Normal EBNM promotes sparsity/distinctness without rotation)
- Taylor refresh defaults to OFF (standard IRLS-within-VI; more stable)
- V1 preserved as-is for reference; V2 is the recommended implementation

Deliverables committed: pushed to `origin/main`

0.11.2 Session 2 (March 5, 2026)

Simulation validation, derivation expansion, and documentation session.

1. **Expanded `REVISED.tex`** from ~650 lines to 1,199 lines with full intermediate algebra throughout every update section:
    - Added new `derivbox` tcolorbox (gray) for algebra steps, distinct from `correctionbox` (red) and `ebnmbox` (blue)
    - **Taylor (Sec 3):** Full 15-step completing-the-square derivation
    - **L update (Sec 4):** Explicit residual substitution, A/B identification for genomics and survival terms separately then combined
    - **F update (Sec 5):** Full quadratic grouping to A_jk, B_jk with R4 fix
    - **beta update (Sec 6):** Variance-mean decomposition E[l^2] = Var(l) + l-bar^2 shown explicitly; error-in-variables correction A_k derived (R5, R6)
    - **tau update (Sec 7):** Full R-bar^2_ij expansion; explicit optimisation step (R7, R8)
    - Recompiled REVISED.pdf: **21 pages** (was 12)
2. **Ran V2.R simulation** (n=250, p=1000, K=5):
    - Key results: RMSE = 0.9978, C-index PCA = 0.827 vs Supervised = 0.828, PH test p = 0.258 (no violation), convergence in 45 iterations
3. **Wrote and rendered `results/simulation_report`** – 12-section comprehensive report, 23 pages via pandoc + xelatex

Deliverables committed: `5da14f3`, `e424d14` (pushed to `origin/main`)

0.11.3 Session 3 (March 10, 2026)

q(beta) derivation, modular implementation, and TDD test suite.

1. **Verified q(beta) derivation** across REVISED.tex (Sec. 6) and March 6 PDF (Sec. 6A):
    - Both agree: A_k = sum_i W_ii * E_q[l^2_ik], B_k = sum_i W_ii * z^{-k}_i * l-bar_ik
    - Confirmed March 6 PDF Eq. 68 typo is intermediate-step only; flagged as R5 in REVISED
    - Confirmed V2.R omits 1/sigma^2 prior term intentionally (EBNM handles it empirically)
    - Proved z_no_k reuse is mathematically valid
    - Confirmed: s_k = 1/sqrt(A_k) is SMALLER when A_k is larger (not larger)
2. **Created test infrastructure** (`tests/` directory):
    - `tests/test_helpers.R`: assert_near, assert_true, assert_equal, assert_length, etc.
    - `tests/run_tests.R`: master runner; exits status 1 if any tests fail
3. **TDD Red phase**: 24 failing tests first (9 groups: math identities, WLS limit, K=1 recovery, K=5 recovery, null shrinkage, error-in-variables, numerical stability, Gauss-Seidel ordering, V2 consistency)

4. **TDD Green phase**: Implemented `code/update_beta.R`:
    - `compute_z_no_k(z, EL, EBeta, k)` – partial working response
    - `update_beta_k(w, z_no_k, EL_k, EL2_k, ...)` – single-factor update
    - `update_beta_all(w, z, EL, EL2, EBeta, ...)` – full Gauss-Seidel loop
5. **Fixed 2 test logic errors** during green phase:
    - T4.1: `length()` returns integer (5L != 5); fixed to `assert_length()`
    - T6.1: s_high should be < s_low (not >); larger A_k -> smaller s_k
6. **Created `derivations/qB/qBeta_update_derivation.tex`** (11 pages):
    - tcolorbox styles: derivbox (gray), correctionbox (red), ebnmbox (blue), verifybox (green), codebox (orange)

Key decisions:

- Plain R test framework (no testthat/DESCRIPTION) – no external dependencies
- `update_beta_k` decoupled from CAVI loop; returns diagnostic fields (A, B, x, s)
- Gauss-Seidel propagation: EBeta_curr[k] updated immediately within the loop

---

0.11.4 Session 4 (March 11, 2026)

Demo for beta update + to-do setup for L/F/tau modules.

1. Created `demos/demo_update_beta.R` – 5 interactive scenarios: anatomy of A_k/B_k, signal recovery K=1, sparsity/shrinkage, error-in-variables, multi-factor K=5. Format: sep/sec/val helpers, pure narrative output (no PASS/FAIL).
2. Updated `PROJECT_STATUS.md` to add `demos/` to the repo map.
3. Updated `code/SupervisedMF_Context.md` with expanded L/F/tau next-step details.

Deliverables committed: `ad80c4e` (pushed to `origin/main`)

0.11.5 Session 5 (March 12, 2026)

Full implementation of modular updates for q(L), q(F), and q(tau).

1. **`code/update_L.R`** – Three functions:
    - `compute_R_k(Y, EL, EF, k)`: Partial residual matrix (n x p), shared with F update
    - `update_L_k(...)`: Vector EBNM – A_L[i] and B_L[i] are n-vectors (patient-specific Cox weights W_ii); two additive sources (genomics + survival)
    - `update_L_all(...)`: Gauss-Seidel loop over K factors

2. **code/update_F.R** – Two functions:
    - update_F_k(...): Pure genomics EBNM – tau_j **cancels** in x_j = B_F[j]/A_F[j] but does NOT cancel in s_j
    - update_F_all(...): Gauss-Seidel loop over K factors
3. **code/update_tau.R** – Three functions:
    - compute_var_term(EL, EL2, EF, EF2): Variance correction (EL2 x EF2' - EL^2 x EF'^2)
    - compute_expected_residual_sq(...): R2_bar = naive residual^2 + Var_Term
    - update_tau(...): Closed-form MLE tau-hat_j = n / colSums(R2_bar) + ELBO proxy
4. **Tests** (9 groups each, all passing):
    - 28 tests for update_L.R (T9 verifies bit-for-bit match with V2.R lines 310–321)
    - 26 tests for update_F.R (T1.4/T1.5 verify tau cancellation numerically)
    - 27 tests for update_tau.R (T4 verifies variance correction direction)
    - Master runner: 105/105 total tests passing
5. **Demos** (5 scenarios each) and **LaTeX derivations** (compiled to PDF) for all three modules.

Key design decisions:

- compute_R_k shared between update_L.R and update_F.R (F sources it from update_L.R)
- tau does not appear in the survival term – F and tau updates are pure genomics
- update_tau has NO per-k loop and NO ebnm dependency (only base R)
- ELBO proxy computed inside update_tau as a convergence monitor

---

0.11.6 Session 6 (March 20, 2026)

Comprehensive documentation: companion docs, roxygen2, and inline comments.

1. **Created docs/ directory** with companion documentation for all four modules:
    - docs/update_beta.qmd – Overview, math background, function reference, test + demo explanations
    - docs/update_L.qmd – Dual-source precision/signal, compute_R_k shared dependency
    - docs/update_F.qmd – tau cancellation property, pure genomics
    - docs/update_tau.qmd – Closed-form MLE, variance correction, ELBO proxy
2. **Created docs/Makefile**: make all renders all 4 .md files to PDF + HTML (Helvetica/Menlo fonts for Unicode Greek letter support).
3. **Added roxygen2 package-readiness tags** to all 11 functions across 4 R scripts:
    - @export, @family (beta_update / L_update / F_update / tau_update), @seealso
4. **Added targeted inline comments** (~3–5 per file) explaining *why* at non-obvious points.

5. **Verification:** 105/105 tests still passing; all 8 rendered outputs (4 PDF + 4 HTML) confirmed.

0.11.7 Session 7 (March 20, 2026)

Modular simulation report, results/ reorganization, and Quarto conversion.

1. **Reorganized `results/`**:
   - Created `figures/{full_sim,modular_sim}/` and `tables/{full_sim,modular_sim}/`
   - Moved existing V2 figures/CSVs into `full_sim/` subfolders
   - Updated `run_simulation.R` output paths accordingly
2. **Created `results/run_modular_simulation.R`** (~340 lines):
   - Completely standalone – sources only the 4 modular update modules
   - `calc_cox_taylor()` copied verbatim from V2.R; no other V2 dependency
   - Block-by-type update ordering: all L -> all F -> all beta -> tau (vs. V2's by-factor ordering; equivalent CAVI objective)
   - Results: RMSE=0.9979, C-index=0.858, PH test p=0.44, null factor beta correctly zeroed
3. **Created `results/modular_sim_report.qmd`** (863 lines):
   - Self-contained Quarto PDF; CAVI runs inline with `#| cache: true`
   - ggplot2 figures (theme_bw), kableExtra tables, Quarto callout blocks
   - Per-figure commentary (~150–300 words each) with statistical insights
4. **Created `results/simulation_report.qmd`** (786 lines):
   - Sources V2.R inline with `pdf(NULL)` + `sink()` wrappers to suppress output
5. Deleted `.md` report files; `.qmd` + `.pdf` are the canonical report artifacts.

Deliverables committed: `9a59c3e` (pushed to `origin/main`)

0.11.8 Session 8 (March 20, 2026)

Documentation housekeeping: CLAUDE.md, file role clarification, PROJECT_STATUS to Quarto.

1. **Created `CLAUDE.md`** – thin Claude Code entry point (key instructions + quick-reference table); defers to `PROJECT_STATUS` for full context.
2. **Discussed file role separation:**
   - `README.md` = GitHub front page (overview, quickstart, brief math)

- PROJECT_STATUS.qmd/.pdf = full internal working document (session log, derivation review, algorithm design decisions, next steps)
- CLAUDE.md = lean Claude Code instructions pointing to PROJECT_STATUS

3. **Converted PROJECT_STATUS.md to PROJECT_STATUS.qmd** for PDF rendering:
   - Added Quarto YAML header (Helvetica Neue / Menlo, TOC, numbered sections)
   - Added callout blocks for key notices (derivation errors warning, V2 recommended tip, test suite status note)
   - Deleted PROJECT_STATUS.md; .qmd + .pdf are now canonical
4. Updated MEMORY.md documentation convention to reflect the three-file structure.

Deliverables committed: pushed to origin/main

0.11.9 Session 9 (March 20–31, 2026)

Factor-wise modular CAVI: fit_modular.R, synthetic report reorganization, legacy code cleanup.

1. **Created code/fit_modular.R** (fit_supervised_mf_modular()):
   - Factor-wise Gauss-Seidel CAVI: within each iteration, cycle over k = 1..K and update (L_k, F_k, β_k) in sequence before moving to k+1. This is the true block coordinate descent with immediate propagation — distinct from the earlier block-wise modular runner which updated all-L, then all-F, then all-β.
   - Internally sources update_L.R, update_F.R, update_beta.R, update_tau.R.
   - Returns list(EL, EL2, EF, EF2, EBeta, EBeta2, Tau, history).
   - history$converged, history$n_iter, history$rmse, history$elbo_proxy.
2. **Created results/modular_sim_factor/run_factor_modular_simulation.R**:
   - Comprehensive analytics runner sourcing fit_modular.R.
   - Produces 8 figure pairs (PDF+PNG), 11 CSV tables for synthetic runs.
   - Designed for both synthetic and real-data modes (real-data support added fully in Session 10).
3. **Created docs/fit_modular.qmd/.pdf/.html**:
   - Full code walkthrough for fit_modular.R, covering factor-wise loop structure, Taylor linearization, convergence criterion, and comparison with block-wise runner.
4. **Reorganized results/** into per-simulation subdirectories:
   - results/modular_sim_factor/synthetic/ — factor-wise CAVI report (canonical)
   - results/modular_sim_block/ — block-wise modular (deprecated)
   - results/full_sim/ — V2 monolithic
   - results/tables/synthetic/ and results/figures/synthetic/ for canonical outputs

5. **Moved legacy code** to `code/legacy/`:
    - `code/legacy/Supervised_Bayesian_MF.R` — V1 original
    - Updated all references in CLAUDE.md, README.md, and runner scripts.
6. **Simulation results (synthetic, factor-wise, n=250, p=1000, K=5, seed=42):**
    - RMSE = 0.9979 (converges to true noise SD)
    - ELBO proxy: non-decreasing
    - β signs: (–, +, 0, –, +) — correct sign recovery, Factor 3 (β=0) zeroed
    - C-index: PCA 0.827 vs Supervised ~0.86
    - Converged at iteration 61 (dL < 1e-3, dBeta < 1e-3)

Deliverables committed: `58cda7b`, `dca6fe4`, `110872c`, `0b560db`, `94ed9a2`

0.11.10 Session 10 (March 31, 2026)

Real PDAC data integration: 7 cohorts, pooled RNA-seq, test suite, and PDAC report.

0.11.10.1 10.1 — Runner extension for real data

Extended `results/modular_sim_factor/run_factor_modular_simulation.R` with:

Configuration block (env-var overrides for scripted/Longleaf use):

```
data_mode    <- "synthetic"   # "synthetic" or "real"
dataset_name <- "TCGA_PAAD"  # single-dataset mode
run_all      <- FALSE         # TRUE: loop all 7 PDAC cohorts
top_n_genes  <- 5000          # top-N by variance; NULL = all
K            <- 5
pdac_data_root <- Sys.getenv("PDAC_DATA_ROOT", unset = ...)
```

New helpers: - `filter_top_genes(Y, gene_names, top_n)` — selects top-N most-variable columns; passthrough if `top_n ≥ p` or `top_n = NULL`. Fixed with `drop = FALSE` to prevent silent matrix-to-vector coercion when `top_n = 1`. - `load_real_data(dataset_name, pdac_root, top_n)` — loads via `load_data_internal.R` (symlink trick to handle relative `data/original/` paths), subsets to `keep == 1` samples, transposes ex matrix, extracts gene symbols, applies variance filter, **column-centres Y**. - `pool_datasets(ds_list)` — intersects gene names, aligns columns, row-binds Y, concatenates time/status for horizontal integration.

Multi-dataset loop (when `run_all = TRUE`): - Iterates over all 7 PDAC cohorts; accumulates per-dataset summary rows. - Writes `results/tables/PDAC_cross_dataset/cross_dataset_summary.csv`. - After loop: pools TCGA_PAAD + Dijk + PACA_AU_seq → fits one model → outputs to `results/tables/PDAC_pooled_rnaseq/` and `results/figures/PDAC_pooled_rnaseq/`.

0.11.10.2 10.2 — Critical bug fixes

Three bugs identified and fixed:

Bug	Root cause	Fix
RMSE increasing on real data	Missing column centring	`Y <- sweep(Y, 2, colMeans(Y), "-")` in `load_real_data()`
Dataset column wrong in summary	`run_pipeline()` used global `dataset_name`	`row$Dataset <- ds` after `run_pipeline()` in loop
Filter crash when `top_n = 1`	`Y[, scalar_idx]` drops matrix to vector	`Y[, keep_idx, drop = FALSE]`

Column centring is the most important fix. Without it, SVD initialisation captures the global mean expression in Factor 1; CAVI then fights the mean artefact each iteration, causing RMSE to *increase* rather than decrease. Column centring is standard preprocessing for matrix factorisation on genomics data.

0.11.10.3 10.3 — Convergence results (K=5, top_5000 genes, max_iter=300)

Dataset	Platform	n	Converged	Iterations	Final RMSE
TCGA_PAAD	RNA-seq	144	TRUE	14	~0.99
Dijk	RNA-seq	90	TRUE	~20	~0.99
PACA_AU_seq	RNA-seq	52	TRUE	~30	~0.99
Moffitt_GEO_array	Microarray	123	TRUE	~50	~0.99
PACA_AU_array	Microarray	63	TRUE	~60	~0.99
Puleo_array	Microarray	288	TRUE	~80	~0.99
CPTAC	RNA-seq	129	TRUE	259	~0.99

CPTAC required max_iter = 300 (was 100, then 250); it converged at iteration 259.

0.11.10.4 10.4 — Test suite: `tests/test_real_data_loading.R`

77 tests, auto-skip if `PDAC_DATA_ROOT` not found (CI-safe): - **T1** (6 tests): `filter_top_genes()` — variance selection, passthrough, NULL, single-gene (`drop=FALSE`), dimension consistency - **T2** (63 tests = 7 × 9): all 7 datasets load without error, correct dimensions, no NAs, time > 0, status {0,1}, numeric matrix, row/col alignment, Y is column-centred (colMeans ~ 0 to tolerance 1e-10), gene_names length matches p - **T3** (6 tests): `pool_datasets()` — non-empty intersection, nrow == sum(n), ncol == intersection length, no NAs, dataset_labels length, error on empty intersection - **T4** (2 tests): synthetic regression — RMSE [0.95, 1.05], convergence == TRUE

77/77 passing; 105/105 core module tests still passing.

0.11.10.5 10.5 — PDAC report

`results/modular_sim_factor/PDAC/factor_modular_sim_report_PDAC.qmd` — fully standalone Quarto report rendering to PDF + HTML.

Self-contained model background: - Model equations (Y = LF' + E, Cox PH), EBNM prior, CAVI pseudocode, fit settings table (K=5, max_iter=300, tol=1e-3, top_5000 genes)

"How to Read These Results" section: - Explains all 8 figure types (RMSE trace, ELBO proxy, β estimates, GEP heatmap, KM curves, PH test, C-index comparison, τ distribution) with what/how/why narrative — no need to reference the synthetic report.

Per-dataset sections (7 datasets + pooled fit): - Convergence history table (first 5 + last 5 iterations), β estimates, factor summary table, C-index comparison, PH test, top-10 genes per GEP, τ distribution figure (fig8)

Dataset-specific figure commentary — computed R values emit short italic paragraphs after each figure, e.g.: - RMSE: drop magnitude and iteration count - ELBO: ascending vs non-monotone check - β: count of prognostic vs null factors - Factor summary: Cox β vs log-rank p consistency - GEP heatmap: dominant factor PVE and prognostic status - KM: which factors show significant stratification - PH test: pass/fail with action language - C-index: contextual (supervised wins/comparable/lower vs PCA) - τ: platform-specific interpretation (RNA-seq/Microarray)

Cross-dataset sections: - Executive summary table (all 7 datasets: convergence, RMSE, C-index) - Cross-platform PACA-AU pair comparison (β correlation, top-gene Jaccard) - Within-modality comparisons (RNA-seq trio, microarray trio) - CPTAC standalone section - Pooled RNA-seq section (dataset membership, pooled fit results, batch caveat)

`ds_display()` helper replaces underscores with spaces in all LaTeX caption strings (prevents "Missing $ inserted" from TeX math-mode interpretation).

Renders to PDF (xelatex, via Quarto) and HTML without errors.

0.11.10.6 10.6 — Future directions recorded

- **Prediction task using hold-out test set:** Apply trained model to a held-out subset of each cohort; evaluate C-index on out-of-sample patients. Requires a `predict_supervised_mf()` function to propagate new patients through the learned F and β.
- **K selection via cross-validation:** Held-out log-likelihood or C-index as objective; compute-intensive, planned for Longleaf HPC.
- **Full ELBO tracking:** Currently a genomics-only proxy; survival Taylor term and KL divergences not yet included.
- **Batch-corrected pooled model:** Apply ComBat or similar before pooling RNA-seq cohorts to remove dataset-of-origin effects.

Deliverables committed: `b3e23b5`, pushed to `origin/main` as part of session 11.

0.11.11 Session 11 (March 31, 2026)

Phases 1–7: Batch correction, multi-init, prior threading, hold-out prediction, feature selection, K selection, and report.

0.11.11.1 11.1 — Phase 1: limma batch correction for pooled RNA-seq

Added `apply_batch_correction(Y, batch_labels)` helper to `run_factor_modular_simulation.R`. Uses `limma::removeBatchEffect()` applied after `pool_datasets()` for the pooled RNA-seq fit. `limma` is a Bioconductor package (installed via `BiocManager::install("limma")`). The function transposes Y before calling limma (which expects genes × samples) and transposes back. Controlled by `batch_correct = TRUE` (default) / `BATCH_CORRECT` env var.

0.11.11.2 11.2 — Phases 2 + 3: Multi-init and prior_family threading

Multi-init (`code/fit_modular.R`): - New parameters: `prior_family` and `init_method` added to `fit_supervised_mf_modular()`. - `init_method = "random"` initialises L and F with `rnorm(sd = 0.1 * sd(Y))` (scaled to prevent Cox Taylor overflow at startup). - Multi-init loop in runner: when `n_init > 1` and `init_method = "random"`, runs `n_init` independent restarts and keeps the fit with the highest final ELBO proxy. Results saved to `multi_init_elbos.csv`.

Prior threading: - `prior_family` is now passed through to all three EBNM update calls inside the factor loop: `update_L_k(..., prior_family)`, `update_F_k(..., prior_family)`, `update_beta_k(..., prior_family)`. Previously the argument existed in the signature but was not forwarded (defaulted to `"point_normal"` everywhere).

0.11.11.3 11.3 — Phases 4–6: Hold-out prediction, feature selection, K selection

`code/predict.R` — `predict_supervised_mf(Y_test, EF, EBeta, lambda)`: - Pseudo-inverse projection: $\hat{L}_{\text{test}} = Y_{\text{test}}\hat{F}(\hat{F}^\top\hat{F} + \lambda I)^{-1}$ - Risk score: $\hat{L}_{\text{test}}\hat{\beta}$ - `lambda = 1e-8` ridge regularisation to handle rank-deficient F.

`code/train_test_split.R` — `stratified_split(status, test_frac, seed)`: - Splits events and censored patients separately, each at `test_frac` fraction. - Guarantees ≥ 1 event and ≥ 1 censored in both train and test. - Returns `list(train_idx, test_idx, n_train, n_test)`.

`code/feature_selection.R` — `cox_feature_selection(Y, time, status, p_thresh)`: - Fits one univariate Cox model per gene; returns indices of genes with `p < p_thresh`. Applied to training data only to prevent survival leakage.

`code/select_K.R` — `auto_prune_K()` + `select_K_cv()` stub: - `auto_prune_K(Y, time, status, K_max, beta_thresh, pve_thresh, ...)`: fits with K_max, flags factor k as active if `|β_k| > beta_thresh` OR `PVE_k > pve_thresh`. Returns `K_effective` and the full fit object. - `select_K_cv()`: errors by design with a message pointing to Longleaf HPC.

`tests/test_predict.R`: 19 tests for `predict_supervised_mf()` and `stratified_split()`. Total test count: 105 + 19 = **124/124 passing**.

0.11.11.4 11.4 — Phase 7: Report update, rendering, documentation

PDAC report updated with: - Hold-out C-index section (PN K=5, stratified 80/20, 7 cohorts) - Conclusion table updated with feature status

Runner updated with env-var overrides for all new parameters: `PRIOR_FAMILY`, `N_INIT`, `INIT_METHOD`, `BATCH_CORRECT`, `HOLDOUT_EVAL`, `FEATURE_SELECTION`, `K_SELECT`.

Hold-out results (PN K=5, 80/20 split, seed=42):

Dataset	C Train	C Test (SBMF)	C Test (PCA)
TCGA_PAAD	0.385	0.341	0.327
CPTAC	0.315	0.389	0.371
Dijk	0.383	0.381	0.424
Moffitt_GEO_array	0.429	0.475	0.469
PACA_AU_array	0.353	0.405	0.262
PACA_AU_seq	0.210	0.185	0.178
Puleo_array	0.340	0.396	0.374

Deliverables committed: `89ff1ff`, `ff1a121`, `3262821`

0.11.12 Session 12 (April 1, 2026)

Prior family comparison (executed), K auto-prune (executed on all 7 cohorts), PDAC report editorial review.

0.11.12.1 12.1 — Prior family comparison: run_prior_k_comparison.R

New script: `results/modular_sim_factor/run_prior_k_comparison.R`. Standalone (no global-variable dependencies on main runner). For each of the 7 datasets:

1. Reads existing PN K=5 holdout results from CSVs (avoids redundant refit).
2. Fits `point_laplace` K=5 with 80/20 hold-out; saves to `{dataset}_pl/`.
3. Runs `auto_prune_K(K_max=10)` to identify K_effective; saves PVE table and scree figure to `{dataset}/k_selection_pve.csv` and `figK_pve_scree.pdf/png`.
4. Refits with K=K_effective (PN) + hold-out; saves to `{dataset}_Keff/`.
5. Accumulates all results into two cross-dataset summary CSVs.

0.11.12.2 12.2 — Results: prior comparison

`results/tables/PDAC_cross_dataset/prior_comparison.csv`

point_laplace K=5 vs. point_normal K=5 (hold-out C-index):

Dataset	C Test PN K=5	C Test PL K=5	ΔC
TCGA_PAAD	0.341	0.707*	+0.366
CPTAC	0.389	0.605	+0.216
Dijk	0.381	0.483	+0.102
Moffitt_GEO_array	0.475	0.580*	+0.105
PACA_AU_array	0.405	0.667	+0.262
PACA_AU_seq	0.185	0.889	+0.704
Puleo_array	0.396	0.594	+0.198

∗ = hit 300-iteration convergence limit. `point_laplace` outperforms `point_normal` in all 7 cohorts. This is the primary motivating finding for switching the default prior to `point_laplace` in future analyses.

0.11.12.3 12.3 — Results: K auto-prune

`results/tables/PDAC_cross_dataset/k_selection_summary.csv`

Dataset	K_max	K_effective
TCGA_PAAD	10	9
CPTAC	10	8
Dijk	10	**10** (saturated)
Moffitt_GEO_array	10	**10** (saturated)
PACA_AU_array	10	8
PACA_AU_seq	10	9
Puleo_array	10	**10** (saturated)

K=5 was insufficient for every dataset. Three cohorts saturate at K_max=10, suggesting their true dimensionality exceeds 10. Re-running with K_max=15 on Longleaf HPC is recommended for these three.

PN K_eff vs. PN K=5 hold-out C-index improvement was positive for all cohorts (largest: TCGA_PAAD +0.28, PACA_AU_array +0.33, PACA_AU_seq +0.56).

0.11.12.4 12.4 — PDAC report editorial review

`results/modular_sim_factor/PDAC/factor_modular_sim_report_PDAC.qmd` overhauled:

- **Executive Summary:** Expanded callout to cover all four key findings (convergence, in-sample results, prior comparison, K selection).
- **Fit Settings table:** Added notes that K is now data-driven and prior family is compared in later sections.
- **CPTAC section:** Removed duplicated factor summary table; replaced with concise CPTAC spotlight with cross-references to new sections.

- **Prior Comparison section:** Fixed broken sentence fragment; quantified top gains; noted the missing PL + K_eff experiment.
- **K Selection section:** Moved CV runtime note after the tables; interpretation now explicitly names the three saturating cohorts.
- **Conclusion:** Completely rewritten as a narrative covering clinical motivation, what SBMF does, four key findings, six current limitations, and six future directions.

Deliverables committed and pushed: `49dd8f6`, `ab23228`

0.11.12.5 12.5 — Open items / next session priorities

1. ~~Run `point_laplace` at K_`effective` per dataset (the expected best configuration).~~ **Done in Session 13.**
2. Gene set enrichment analysis on top-loaded genes per GEP (fgsea / Enrichr).
3. K selection via CV on Longleaf HPC (`select_K_cv()` stub is ready).
4. Re-run Dijk, Moffitt, Puleo with K_max=15 to probe true dimensionality.
5. ~~Full ELBO tracking (add Cox likelihood and KL terms to ELBO proxy).~~ **Done in Session 13.**

0.11.13 Session 13 (April 1–2, 2026)

Full ELBO tracking implemented; point_laplace at K_eff run for all 7 cohorts; all results re-run with consistent methodology; PDAC report updated.

0.11.13.1 13.1 — Full ELBO implementation

New file: `code/compute_elbo.R`. Provides two functions:

- **`compute_ebnm_kl(ebnm_log_lik, A, x, mean_q, second_q)`** — extracts the KL divergence $E_q[\log g(\theta)] - E_q[\log q(\theta)]$ from one EBNM update by subtracting the expected pseudo-log-likelihood from the returned marginal log-likelihood. Works for vectors (L, F updates) and scalars (β update). Returns a scalar ≤ 0.
- **`compute_survival_elbo(logPL, w, EL, EL2, EBeta, EBeta2)`** — computes $E_q[\log \text{PL}(t, \delta \mid L, \beta)]$ via second-order Taylor approximation: $\log \text{PL}(\eta_0) - \frac{1}{2}\sum_i w_i \cdot \text{Var}_q(\eta_i)$, where $\text{Var}_q(\eta_i) = \sum_k [EL2_{ik} \cdot EBeta2_k - EL_{ik}^2 \cdot EBeta_k^2]$.

`code/fit_modular.R` updated (7 edits): - `calc_cox_taylor()`: returns `$logPL` (Breslow partial log-likelihood at η_0) - `source("code/compute_elbo.R")` added alongside other sources - `history$elbo_full` initialized and populated each iteration - KL accumulators (`kl_L`, `kl_F`, `kl_beta`) filled inside the k-loop from EBNM outputs - Full ELBO = `elbo_proxy + surv_elbo + sum(kl_L) + sum(kl_F) + sum(kl_beta)` - Verbose print shows both ELBOs; convergence trim extended to `elbo_full`

`tests/test_elbo.R` (new, 15 tests):

- Section 1 (4 tests): `calc_cox_taylor` backward compat + new `$logPL` field
- Section 2 (3 tests): `compute_ebnm_kl` — scalar output, scalar inputs, KL ≤ 0
- Section 3 (3 tests): `compute_survival_elbo` — scalar output, zero-variance = logPL exactly, uncertainty reduces value
- Section 4 (5 tests): integration via `fit_supervised_mf_modular` — `elbo_full` present, finite, all-iters tracking

`Rscript tests/run_tests.R`: **139/139 passing** (24 β + 28 L + 26 F + 27 τ + 19 predict + 15 elbo).

0.11.13.2 13.2 — Complete 4-condition prior comparison

`run_prior_k_comparison.R` updated to run all 4 conditions for all 7 datasets:

1. point_normal K=5 (PN_K5)
2. point_laplace K=5 (PL_K5)
3. point_normal K_eff (PN_Keff)
4. point_laplace K_eff (PL_Keff) — **new**

All conditions re-run with consistent methodology (elbo_full as `ELBO_Final`). New `{dataset}_pl_Keff/` directories created for all 7 datasets with `factor_summary_table.csv`, `holdout_cindex.csv`, and `convergence_history.csv` (ELBO_Proxy + ELBO_Full columns).

Key result — PL K_eff vs. PN K_eff:

Dataset	PN K_eff C_Test	PL K_eff C_Test	Winner
TCGA_PAAD	0.625	0.620	PN
CPTAC	0.635	0.623	PN
Dijk	0.544	0.510	PN
Moffitt_GEO_array	0.530	0.550	PL
PACA_AU_array	0.738	0.738	Tie
PACA_AU_seq	0.741	0.778	PL
Puleo_array	0.674	0.649	PN

Conclusion: The dominant driver of improvement is K selection (K_eff vs K=5), not prior family. At K_eff, `point_normal` wins in 4/7 cohorts, `point_laplace` wins in 2/7, tie in 1/7. The advantage of `point_laplace` is strongest when K is fixed at 5.

0.11.13.3 13.3 — PDAC report updates

- Fixed `$+$` LaTeX artifact in Key Findings (was rendering as literal `$+$0.012`)
- Prior Family section: updated intro to mention 4 conditions; table extended to 4 conditions with new δ(PL K_eff − PN K=5) column; plot extended with 4th condition (purple diamond); interpretation updated with actual PL K_eff results
- New "Full ELBO vs. Proxy" subsection with convergence comparison plot (TCGA_PAAD)
- Key Findings "Prior family": updated to reflect mixed PL K_eff result

- Limitation 6: replaced "prior comparison incomplete" with "ELBO not yet used for model selection"
- Future Directions: removed completed items (PL K_eff, full ELBO); renumbered remaining
- Re-rendered: both PDF and HTML regenerated successfully

0.11.13.4 13.4 — Open items / next session priorities

1. Gene set enrichment analysis on top-loaded genes per GEP (fgsea / Enrichr).
2. K selection via CV on Longleaf HPC (`select_K_cv()` stub is ready).
3. Re-run Dijk, Moffitt, Puleo with K_max=15 to probe true dimensionality.
4. ELBO-based model selection: use `elbo_full` as criterion for K selection and prior comparison.

0.11.14 Session 14 (April 5–8, 2026)

Lab meeting preparation session.

0.11.14.1 14.1 — Slide deck creation and revision

1. Created V1 slide deck (`presentation/walther_lab_meeting_04_09_2026/V1/lab_meeting_april9_` — initial draft covering model motivation, algorithm, synthetic validation, and PDAC results.
2. Created V2 slide deck with revised slide text, tighter layout, and updated PDAC figures (`presentation/walther_lab_meeting_04_09_2026/V2/lab_meeting_april9_V2.qmd`).
3. Added dataset description table on the PDAC cohorts slide to orient the audience.

0.11.14.2 14.2 — Slide-text corrections

Fixed 4 factual discrepancies identified during review:

1. CPTAC iteration count — corrected to match the value in the convergence table.
2. RMSE direction — clarified that lower RMSE is better; phrasing was ambiguous.
3. Puleo factor labels — aligned slide labels with the factor summary table output.
4. TCGA KM language — softened "significant" to match actual PH test p-value.

0.11.14.3 14.3 — Companion document and repository hygiene

1. Created companion LaTeX document (`presentation/walther_lab_meeting_04_09_2026/Notes/lab_` — 21-page prep guide covering figure analysis, Q&A preparation, and review sections for each slide.
2. Fixed `.gitignore`: removed the `**/*.tex` rule that was silently blocking `companion.tex` from being tracked; added `**/*.out` to suppress LaTeX auxiliary output files.
3. Reorganized the presentation directory into V1/V2/Notes subdirectories for cleaner version management.

0.11.15 Session 15 (April 9, 2026)

Post-lab-meeting fixes and feedback processing.

0.11.15.1 15.1 — Lab meeting presentation

1. Presented at Rashid lab meeting; received structured feedback from Naima Rashid and Tyler Humphreys covering model clarity, C-index computation, code organization, and research directions.

0.11.15.2 15.2 — Figure 3 permutation bug fix

1. Fixed a non-bijective permutation bug in Figure 3 (synthetic β̂ comparison): `apply(abs(cors), 2, which.max)` was used to align estimated factors to true factors, but this can assign the same true factor to multiple estimated factors when two estimated factors are most correlated with the same true factor.
2. Replaced with a `bijective_match()` helper implementing greedy global-maximum assignment: iteratively selects the (estimated, true) pair with the largest absolute correlation, assigns it, removes both from consideration, and repeats — guaranteeing a one-to-one mapping.

0.11.15.3 15.3 — Synthetic simulation seed fix

1. Changed synthetic simulation seed from 42 → 222: seed 42 produced a data configuration where True F1 and True F5 (the null factor) were mixed after permutation alignment, causing the model to assign β̂_null = −0.77 to the null factor — an artifact of the seed, not the model.
2. Seed 222 cleanly separates all 5 factors and correctly zeroes the null factor (β̂ ≈ 0), giving a cleaner validation of the supervised component.

0.11.15.4 15.4 — Companion document update

1. Updated companion doc V2 (`presentation/walther_lab_meeting_04_09_2026/Notes/lab_meeting`) to reflect the corrected Figure 3 permutation alignment and seed change.

0.11.15.5 15.5 — Open items / next session priorities

1. V3 slide deck — targeted clarifications on slides covering model specification, objective function, EBNM, CAVI, and synthetic validation, in response to lab meeting feedback.
2. C-index code fix — use the model's own β (i.e., `EL %*% EBeta`) to form the linear predictor instead of refitting a new `coxph` on training data; this tests the model's survival predictions rather than a post-hoc re-fitted Cox model.
3. Create `DECISIONS.md` — 12 architectural decision entries documenting key modelling and implementation choices made between Feb 12 and Apr 9 (e.g., Gauss-Seidel ordering, orthogonalization off, Taylor approximation for survival ELBO).
4. Create `ROADMAP.md` — research directions: incorporating F'Y term in survival model (R1), adding λ scaling parameter (R2), cross-cohort validation (R3), inter-dataset normalization strategy (R4), K selection strategy (R5), `{targets}` workflow for reproducibility (R6).
5. Create `config/globals.yml` — centralize hard-coded parameters (convergence thresholds, K_max defaults, simulation seeds) currently scattered across `fit_modular.R`, `select_K.R`, and the simulation runner.
6. Repo architecture review — propose a cleaner directory layout without moving files yet; surface ambiguities in current structure for discussion.
7. Re-render synthetic and PDAC simulation reports after the C-index fix to update all reported C-index values.

0.11.16 Session 16 (April 23–24, 2026)

DeSurv benchmark implementation (Phases 0–3B of `plans/DeSurv_Benchmark_4_23_26.md`). Initial Codex implementation (Phases 0–3A) followed by continued development in this session.

0.11.16.1 16.1 — DeSurv-aligned preprocessing

1. Added `code/preprocess_desurv.R` with five helper functions: `log2_plus1_transform()`, `select_top_variable_genes()`, `rank_transform_subjects()`, `preprocess_desurv_cohort()`, `intersect_preprocessed_cohorts()`, `merge_preprocessed_cohorts()`.
2. Pipeline matches DeSurv (Young et al. 2025): $\log_2$(counts+1) for RNA-seq top-2000 variable genes rank-transform per subject. Inputs supplied on a log scale (microarray, CPTAC) skip the $\log_2$ step.

0.11.16.2 16.2 — Alpha CV selection

1. Added `code/select_alpha_cv.R` with `select_alpha_cv()`: stratified K-fold CV over a grid of α values, returns optimal α via 1-SE rule.
2. `config/globals.yml` alpha_grid expanded from [0.1, 0.3, 0.5, 0.7, 0.9] to [0.0, 0.1, 0.3, 0.5, 0.7, 0.9, 1.0] to include boundary conditions.

0.11.16.3 16.3 — predict.R SVD pseudoinverse fix

1. `predict_supervised_mf()` in `code/predict.R` replaced `solve(crossprod(EF) + lambda*I)` with SVD-based Moore-Penrose pseudoinverse. Fixes crash when ARD collapses factor columns during early CV iterations (near-singular Gram matrix).

0.11.16.4 16.4 — Benchmark runner and outputs

1. Scripts moved: `run_ssbmf_benchmark.R` and `ssbmf_summary_report.qmd` now live in `results/benchmark_sim/`. Old outputs archived to `results/modular_sim_factor/archive/`.
2. `run_ssbmf_benchmark.R` contains both synthetic benchmark (`run_ssbmf_benchmark()`) and PDAC cross-cohort benchmark (`run_real_data_benchmark()`).
3. Outputs written to `results/benchmark_sim/outputs/synthetic/` and `outputs/real_data/`.

0.11.16.5 16.5 — Synthetic DGP fix

1. Removed 5× amplitude multiplier on null factors in `generate_synthetic_benchmark_data()`. Equal amplitudes across all factors ensure PCA does not incidentally dominate via null factors.
2. `beta_true` now read from `cfg$synthetic$b_true = [1.5, −1.2, 0.8, −0.5, 0.0]` (4 prognostic
    - 1 null) rather than the old hardcoded [1.0, -0.8, 0, 0, 0].
3. Post-fix synthetic results: supervised C-index 0.79 > PCA 0.76; converged in 13 iterations at alpha_opt=0.5; K_effective=4/10; 3/4 prognostic factors recovered.

0.11.16.6 16.6 — Figure expansion (synthetic)

Added 6 new figure types to the synthetic benchmark (total: 11 figures per run): - `rmse_trace`: Reconstruction RMSE over iterations - `pve_scree`: PVE scree with active/pruned colouring - `gep_loading_heatmap`: Top-gene loading heatmap for matched factors - `km_3group_training`: KM curves stratified by linear predictor tertile - `alpha_cv_curve`: Mean CV C-index vs. alpha with 1-SE selected value - `tau_distribution`: Gene precision histogram vs. true τ mean

0.11.16.7 16.7 — PDAC gene intersection and real-data benchmark

1. TCGA_PAAD (n=144, RNA-seq) × CPTAC (n=129, RNA-seq): 838 shared genes after top-2000 filtering. Combined n_train=273, event rates ~52% (TCGA) / 50% (CPTAC).
2. 5-fold CV over 7 alpha values running on PDAC data. External cohort projections to 5 cohorts (Dijk, Moffitt_GEO_array, PACA_AU_array, PACA_AU_seq, Puleo_array) to follow.
3. Summary report: `results/benchmark_sim/ssbmf_summary_report.qmd` → PDF + HTML.

0.11.16.8 16.8 — Test suite

Tests still passing: `Rscript tests/run_tests.R` — expected 171/171 (unchanged from prior session).

0.11.17 Session 17 (April 24, 2026)

Report polish and prior sensitivity comparison for the DeSurv benchmark report.

0.11.17.1 17.1 — Prior sensitivity: point_normal vs. point_laplace

1. Added `prior_beta` parameter to `run_ssbmf_benchmark()` and `run_real_data_benchmark()` in `results/benchmark_sim/run_ssbmf_benchmark.R`. Both now accept `prior_beta = "point_normal"` or `"point_laplace"` and write outputs to subdirectories keyed by prior name.
2. Ran all four PDAC benchmark combinations: `tcga_only/{point_normal, point_laplace}` and `cptac_only/{point_normal, point_laplace}`. Results written to `results/benchmark_sim/out`
3. Key finding: `point_normal` matches or slightly outperforms `point_laplace` on external C-index (TCGA-only: 0.602 pn vs. 0.579 pl; CPTAC-only: 0.628 vs. 0.620). `point_normal` recommended as default.
4. Multi-modal (TCGA+CPTAC merged): all β̂_k = 0 in both priors — ARD correctly diagnoses platform-batch dominance. Documented in Section 6 of the report.

0.11.17.2 17.2 — Report overhaul (ssbmf_summary_report.qmd)

Major rewrite of `results/benchmark_sim/ssbmf_summary_report.qmd` across three rounds:

Content additions: - Executive Key Findings summary section - ARD vs. ELBO grid search justification in Prior Specification (Section 1.3): single-pass efficiency + Bayesian coherence (soft vs. discrete model selection) - Interpretive **Takeaway** paragraph before every figure - Gradient notation for α mixing (∇L_genomics + ∇L_survival) replacing confusing A/B notation - EBeta and EBeta_SD defined explicitly in Methods - Prior sensitivity comparison (Section 5) with both synthetic and PDAC results - Multi-modal failure documented (Section 6) with path-forward (shared-L extension) - point_laplace over-shrinkage noted in beta recovery caption

Formatting fixes: - `\floatplacement{figure}{H}` + `\floatplacement{figure}{H}` □ figures stay inline - Removed all `\FloatBarrier` (redundant with H placement, caused blank gaps) - Removed ––– horizontal dividers (added LaTeX vertical space) - `linestretch` 1.2 □ 1.1; margins tightened to 0.85in/1in - Global figure width 90% □ 72%; diagnostic plots (ELBO, RMSE, alpha-CV, tau) □ 60%; KM figures □ 62%; holdout bar □ 45% - `\newpage` only at top-level section boundaries (removed from within sections) - Key Findings flows after TOC (no forced page break) - Prior comparison figures use Quarto `layout-ncol: 2` for true side-by-side layout - Collapsed double blank lines after `\newpage`

Final page count: 24 pages (reduced from original ~34).

0.11.17.3 17.3 — Output structure reorganisation

- Outputs restructured to `outputs/real_data/{training_mode}/{prior_beta}/` hierarchy
- `training_mode_prior_comparison.csv` written summarising all mode × prior combinations
- Old flat `outputs/real_data/` structure preserved for backwards compatibility

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0.11.18 Session 19 (April 29, 2026)

Multi-platform merged training diagnostics: v2 preprocessing, lambda sweep, EBMF diagnostic, and EBMF warm-start. Root cause of β=0 failure on merged TCGA_PAAD + CPTAC identified.

0.11.18.1 19.1 — v2 Preprocessing (intersect-first pipeline)

Root cause of the 838-gene problem diagnosed: v1 preprocessing applied per-cohort top-2000 selection *before* intersecting gene universes, leaving only ~838 common genes. New v2 pipeline added to `code/preprocess_desurv.R` (`preprocess_merged_cohorts()`):

1. Intersect raw gene universes across all training cohorts (~17,900 genes for TCGA+CPTAC)
2. $\log_2(x+1)$ transform [TCGA only; CPTAC is supplied log2-transformed]
3. Quantile normalization across all merged samples (`preprocessCore::normalize.quantiles()`)
4. Select top-2,000 genes by merged-matrix variance
5. Per-subject rank transform (optional; toggled via `rank_transform` parameter)

Gated by `preprocessing_version = "v2"` in `run_real_data_benchmark()`. v1 pipeline untouched; both coexist. External cohorts still use v1 single-cohort preprocessing.

0.11.18.2 19.2 — Lambda sweep ($\{1,5,10,20\} \times 3$ priors, merged v2)

Script: `results/benchmark_sim/run_lambda_sweep.R` **Results:** `results/benchmark_sim/outputs/real`

Motivated by p/n ≈ 10 scale imbalance between genomics and survival objectives. Tested all 12 combinations on the merged v2 training set (n=273, p=2000).

Key findings:

- β = 0 for **all 12 combinations** — lambda tuning does not help
- λ ≥ 5 causes a new failure: the **entire EL matrix collapses to zero** (not just β), making the model fully degenerate. Amplifying the survival gradient overshoots and destabilizes the loading updates
- Only λ=5 point_laplace showed max|β|=0.032, still below the active threshold of 0.05
- **Decision:** λ=1 retained as default; lambda tuning ruled out for merged training

0.11.18.3 19.3 — EBMF diagnostic (data vs. model problem)

Script: `results/benchmark_sim/run_ebmf_diagnostic.R` **Report:** `results/benchmark_sim/outputs/e`

Ran `flashier::flash()` (K=20, var_type=2) and PCA (K=20) on the same merged v2 training matrix used by SSBMF, with no survival objective. Tested each factor for Cox association.

Key findings:

- **5/20 EBMF factors** Cox-significant ($p < 0.05$); strongest: EBMF6 C-index=0.629, $p=3\times10^{-6}$
- **4/20 PCA components** also significant — consistent evidence across methods
- **Conclusion: survival signal EXISTS in the merged data** □ SSBMF's β=0 is a model problem, not a data problem

Top gene analysis: EBMF3/17 = exocrine pancreas markers (PTF1A, GUCA1C); EBMF4 = B cell/immune (FCRL1, TCL1A, CR2); EBMF6 = metabolic (A2ML1, CYP24A1).

0.11.18.4 19.4 — EBMF warm-start (root cause localization)

Script: `results/benchmark_sim/run_ebmf_warmstart.R`

Two experiments to distinguish β-update bug from initialization/L-update bug.

Experiment 1 — β-only, EL fixed at EBMF loadings: - β moved off zero at iteration 1; converged to 6/20 active factors by iteration 7 - Active: EBMF3/4/6/13/16/17 — exactly matching the Cox-significant factors - **Conclusion: the β CAVI update is not broken**

Experiment 2 — Full CAVI warm-start from EBMF init: - β collapsed back to near-zero (max|β|=0.026) in 23 iterations - L/F updates washed out the EBMF initialization despite it being survival-informative - **Conclusion: the L update is the root cause**

Root cause: In `update_L_k()`, the survival precision term $A_{surv} = \lambda \cdot W_{ii} \cdot E[\beta_k^2]$ is structurally negligible when EBeta ≈ 0 at initialization (A_{gen} sums over p=2000 genes and dominates). The update converges to genomics-reconstruction-optimal factors where β has nothing to select. This is a chicken-and-egg cycle that λ tuning cannot break.

`fit_modular.R` extended with `EL_init` / `EF_init` parameters (init_method="custom") for injecting external initializations. 171/171 tests still passing.

0.11.18.5 19.5 — Documentation

- `DECISIONS.md`: four new entries (v2 preprocessing, lambda collapse, EBMF diagnostic, normal prior ruled out, warm-start root cause localization)
- `ROADMAP.md`: status updated, inter-dataset normalisation marked complete, EBMF warm-start marked complete, new "Debug update_L_k()" item added as next priority
- `docs/update_L_fix.md` (new): full debugging guide for the L-update fix — mechanism, four candidate fixes (reorder inner loop, β burn-in, ridge Cox init, A_surv/A_gen normalization), recommended investigation sequence, success criteria

0.11.18.6 19.6 — Current branch status

All session 19 work lives on branch `preprocessing-reorder`. Ready to merge to `main` — all code changes are backward-compatible, 171/171 tests pass, and the L-update fix is scoped to a separate new branch (`fix-L-update-beta-cycle` or similar).

0.11.19 Session 18 (April 24, 2026)

PH diagnostics, training-mode × prior dot plot, and λ scaling sandbox evaluation.

0.11.19.1 18.1 — Proportional hazards diagnostics

1. Created `results/benchmark_sim/compute_ph_diagnostics.R`: standalone re-fit at saved `alpha_opt=0.5` (TCGA-only, point_normal), runs `cox.zph()` on all 5 external cohorts, saves `ph_diagnostics_table.csv` to `outputs/real_data/tcga_only/point_normal/tables/`.
2. Results: Dijk p=0.77, Moffitt p=0.72, PACA-AU array p=0.34, PACA-AU seq p=0.41 — all PASS. Puleo_array p=0.026 — marginal FLAG, likely a power artefact at n=288.
3. `cox.zph()` also wired into `run_real_data_benchmark()` external cohort loop for future runs.
4. Added Section 4.3 (PH Diagnostics) to `ssbmf_summary_report.qmd`. PH item removed from Open Issues.
5. `DECISIONS.md` entry added (2026-04-24).

0.11.19.2 18.2 — Training mode × prior dot plot

Added Section 5.3 to `ssbmf_summary_report.qmd`: per-cohort C-index dot plot for all four conditions (TCGA/PN, TCGA/PL, CPTAC/PN, CPTAC/PL) × 5 external cohorts. Uses data already loaded in the report setup chunk — no new outputs required.

0.11.19.3 18.3 — Lambda ( ) survival-scaling sandbox

1. Added `lambda = 1.0` parameter to `update_L_k()`, `update_L_all()`, and `fit_supervised_mf_modula` λ scales both `A_surv` and `B_surv` in the L update. Default 1.0 is backward-compatible; all 171 tests pass unchanged.
2. Created `results/benchmark_sim/sandbox_lambda_test.R`: tests λ□{1, p/n=5, 2p/n=10} on the synthetic benchmark (n=250, p=1000, K=5, seed=222).
3. **Results:** hold-out C-index flat at ≈0.805 across all three conditions. β RMSE *worse* at λ=p/n (+0.25) and λ=2p/n (+0.43) than λ=1. β scale error dominated by L–β scale indeterminacy, not gradient imbalance.
4. **Decision:** λ=1.0 retained. Parameter kept exposed for future experimentation.
5. Added `cavi.lambda: 1.0` to `config/globals.yml` (single source of truth).
6. Wired `lambda = cfg$cavi$lambda` through all cfg-aware call sites: `run_ssbmf_benchmark.R` (3 fit calls + 2 function signatures + 4 entry-point call sites) and `run_factor_modular_simulation.R` (3 fit calls).

7. DECISIONS.md entry added (2026-04-24). ROADMAP.md λ item marked complete.

0.11.19.4 18.4 — Results directory cleanup (to-do noted)

results/ mixes scripts, CSVs, figures, and reports in the same subdirectories. A dedicated refactor commit is needed to reorganise into scripts/, reports/, and outputs/ subdirectories. Tracked in ROADMAP.md under Infrastructure.

0.11.20 Session 20 (April 29, 2026)

Cluster A implementation on branch fix-L-update-beta-cycle. Resolves training-side β=0 failure on merged TCGA_PAAD + CPTAC; external generalization is mixed.

0.11.20.1 20.1 — Branch + design doc

- Created branch fix-L-update-beta-cycle from main.
- Committed docs/beta_zero_fix_design.md (420 lines) as the source-of-truth for Cluster A / B / C: diagnostic chain, root-cause analysis (chicken-and-egg cycle + scale imbalance), five-phase plan with explicit session boundaries.

0.11.20.2 20.2 — Code changes (code/fit_modular.R, code/update_L.R)

1. **Instrumentation** (fit_modular.R): logs Cox warm-start EBeta range and per-factor A_gen / A_surv / ratio at iter 1 (k ≤ 3). Confirmed structural imbalance: ratios at iter 1 are 0, 0, 2e-4 for k = 1, 2, 3 — survival precision is ~5000× smaller than genomics.
2. **Fix 1 — inner-loop reorder** β □ L □ F (was L □ F □ β). Justified by symmetric z_no_k / R_k invariance in the current k. Same number of compute_z_no_k() and compute_R_k() calls; no extra computation. Now canonical (no opt-in flag).
3. **Fix 2 — N_burnin parameter** (default 0): pre-CAVI loop runs N_burnin iterations of β-only updates with EL fixed at SVD init. Replicates Warm-start Exp 1 to seed non-zero β before joint CAVI begins.
4. **A2 — alpha_schedule parameter** (default NULL): list(warmup_iters, ramp_iters) ramps α from 0 to target over the warmup+ramp window. Curriculum lets L settle into reconstruction-meaningful directions before survival pressure is applied.
5. **Fix 4 — normalize_AB parameter** in update_L_k (default FALSE): rescales A_surv (and B_surv consistently) up to match A_gen's magnitude. **Reformulated** from docs/beta_zero_fix_design.md §4.8 — the original "divide both by their means" formula collapsed L to zero in the smoke fit (verified: inflated 1/√A_L noise scale, over-shrunk EBNM). The retained reformulation preserves the original EBNM noise interpretation.
6. **All four threaded** through fit_supervised_mf_modular() with backward-compatible defaults.

0.11.20.3 20.3 — Smoke fit on merged TCGA+CPTAC v2 (Step 5 / decision gate)

- Configuration: n=273, p=2000, K=20, alpha=0.5, lambda=1.0, prior_LF=`point_normal`, prior_beta=`point_normal`, N_burnin=10, normalize_AB=TRUE, max_iter=60.
- Cox warm-start EBeta range: [-0.048, 0.061].
- Post-burn-in EBeta range: [-0.034, 0.046] (point_normal prior shrinks slightly during pure-β fitting; not the failure mode).
- Final EBeta range: **[-0.0588, 0.0580] — 2/20 factors active** (|β| > 0.05); k=4 positively associated, k=6 negatively associated.
- ELBO monotone non-decreasing across 60 iterations; max|EL| = 1.83e3 (well above 0.01).
- Decision gate: **PASS** — proceed to Step 5b.

Without `normalize_AB`, the same fit converged with all factors below the |β| > 0.05 threshold (final range [-0.014, 0.024]), confirming Fix 4 is the load-bearing change.

0.11.20.4 20.4 — External-cohort C-index (Step 5b)

Compared Cluster A (`N_burnin=10, normalize_AB=TRUE`) vs. baseline (`N_burnin=0, normalize_AB=FALSE`, both with the unconditional Fix 1 reorder) on the same merged training set, projected via `predict_supervised_mf()` to 5 external cohorts:

Cohort	Cluster A	Baseline	Δ
Dijk	0.6061	0.6255	−0.019
Moffitt_GEO_array	0.5413	0.5294	**+0.012**
PACA_AU_array	0.5756	0.6358	−0.060
PACA_AU_seq	0.6293	0.6529	−0.024
Puleo_array	0.5501	0.6258	−0.076

Cluster A improves on **1/5** cohorts. Baseline wins on 4/5. The recovered β reflects training-Cox-favored L directions that don't transport. This is the design doc's predicted Cluster B trigger (§3 row "EBeta non-zero but unstable").

0.11.20.5 20.5 — Living docs + commit

- `tests/run_tests.R`: 171/171 passing after every step.
- `DECISIONS.md`: 2026-04-29 Cluster A entry.
- `ROADMAP.md`: marked "Debug update_L_k() A_surv/A_gen imbalance" complete; added "Cluster B — Cox-on-YF reformulation" as new High-priority item.
- `CLAUDE.md`: active debugging target updated from "β=0 on merged" to "Cluster B reformulation (external generalization gap)".
- New diagnostic scripts: `results/benchmark_sim/run_cluster_a_smoke.R`, `run_cluster_a_externa`
  Outputs in `results/benchmark_sim/outputs/cluster_a_smoke/` and `cluster_a_external/`.
- Branch `fix-L-update-beta-cycle` commits: `12b0424` (design doc), `55500fd` (Cluster A implementation + smoke + external).

0.11.20.6 20.6 — Implication for Phase 4 (Cluster B)

The training-side β=0 failure is solved as a *recoverability* problem, but not as a *generalization* problem. The Cox-on-YF reformulation (`docs/beta_zero_fix_design.md` §5) addresses both the cold-start cycle (no A_surv ≈ 0 trap on F) and the train/test mismatch (prediction uses the same Y · F · β̃ formula as training). Phase 4 work begins with derivations in `derivations/qF_supervised/` per design doc §5.5.

0.11.21 Session 21 (May 4–5, 2026)

Cluster B implementation completed and merged; benchmark pipeline consolidated; Phase A–C fixes fully applied to both LB and YFB; full benchmark re-run and 05/05 report rendered.

0.11.21.1 21.1 — Cluster B implementation (`code/fit_cox_on_yf.R`, merged to `main`)

Cox-on-YF reformulation (Cluster B) implemented and merged from branch `cox-on-yf-reformulation`. Core components:

- `code/fit_cox_on_yf.R` — main CAVI loop; linear predictor η = (Y·EF_norm)·β
- `code/predict_cox_on_yf.R` — hold-out projection using stored EF_norms and EBeta
- `code/update_L_surv_YFB.R`, `code/update_F_surv_YFB.R` — YFB-specific update variants
- `derivations/cox_on_YF/` — full parameter derivations (LaTeX)
- Smoke test: 3/3 PASS; synthetic: C-index 0.605 vs PCA 0.471 (3 active factors)

0.11.21.2 21.2 — Benchmark consolidation

Created two canonical benchmark runners (replacing 7+ one-off diagnostic scripts):

- `results/benchmark_sim/run_LB_benchmark.R` — Cluster A; 4 sections: synthetic + PDAC train + external validation + CSV output; tests point_normal vs normal side-by-side
- `results/benchmark_sim/run_YFB_benchmark.R` — Cluster B; same structure
- Archived retired scripts to `results/benchmark_sim/archive/`
- `config/globals.yml` updated with `benchmark` section: K=10, K_merged=20, N_burnin=0, normalize_AB=FALSE, cox_warmstart=FALSE, alpha=0.5, lambda=1.0, beta_threshold=0.001

0.11.21.3 21.3 — Phase A–C fixes (both LB and YFB)

Phase A (confirmed correct): CAVI inner loop is L□F□β (Gauss-Seidel), not β□L□F. 18-unit ELBO gap verified on identical data.

Phase B (YFB only): EF column normalization added to `fit_cox_on_yf.R` and `predict_cox_on_yf.R`. Without normalization ‖EF_k‖ ≈ O(√p), driving A_beta ≈ O(p·n) and collapsing EBeta to zero via spike-and-slab. With unit-norm EF, ZF is O(1)-scale. YFB synthetic result: C rescued from 0.119 □ 0.906 (K_eff=4).

Phase C (both models): Training concordance sign correction implemented in both `fit_cox_on_yf.R` (YFB) and `fit_supervised_mf_modular()` in `fit_modular.R` (LB). After CAVI convergence, η_train = EL·EBeta (LB) or ZF·EBeta (YFB); if C_train < 0.5, negate EBeta. Root cause: pmax(SVD) initialisation discards sign information. Controlled by `sign_correction=TRUE` parameter (default TRUE).

Alpha CV bug fixed: Applying Phase C inside CV folds in `select_alpha_cv.R` caused fold-to-fold sign inconsistency — some folds corrected, others not — making all CV C-indices unreliable. 1-SE rule then selected alpha=1.0 (degenerate), yielding K_eff=0 and C≈0.5 everywhere. Fix: `sign_correction=FALSE` in the `fit_supervised_mf_modular()` call inside `select_alpha_cv.R`. The existing `I(-pred$risk_scores)` negation (pre-Phase-C convention) is retained unchanged.

0.11.21.4 21.4 — LB benchmark re-run results (2026-05-05)

All three LB training modes re-run with Phase C fix. Alpha CV (5-fold, 1-SE) selects alpha=0.50 for all modes. Risk scores saved per-subject to `LB_riskscores_*.csv`.

Mode	n_train	K_eff	alpha	External C-index range
tcga_only	144	2	0.50	0.55–0.66 (Dijk=0.584, PACA_AU_seq=0.656, Puleo=0.615)
cptac_only	129	3	0.50	0.55–0.67 (Dijk=0.624, PACA_AU_seq=0.673, Puleo=0.625)
merged	273	3	0.50	0.51–0.67 (Dijk=0.571, PACA_AU_seq=0.671, Puleo=0.644)

All results above 0.5 and broadly consistent with the 04/29 DeSurv archived baseline (~0.60–0.65). Both point_normal and normal priors give identical results for LB — the spike-and-slab collapses to the same active set as the normal prior at these signal levels.

0.11.21.5 21.5 — YFB benchmark status (2026-05-05)

YFB single-cohort modes with normal prior are functional (C=0.55–0.63, tcga_only). YFB merged still collapses to K_eff=0 — TCGA + CPTAC cohort mixing suppresses A_surv/A_gen to 0.0000–0.0011 at iter 1. The F matrix still captures survival-relevant column structure, yielding external C=0.57–0.64 despite K_eff=0 (beta effectively zero). Cox warm-start (`cox_warmstart=TRUE`) was tested on merged and did not rescue beta.

0.11.21.6 21.6 — 05/05 benchmark report

`docs/reports/ssbmf_summary_report_05_05_26.qmd` rendered to PDF and HTML. Report covers: alpha CV tables, training convergence, LB and YFB bar charts (external C-index by cohort), LB vs YFB vs archived-baseline comparison, full external validation tables for all modes, KM curves (high/low risk stratification) for both models, cox warm-start diagnostic table, and archived baseline comparison. Commit: `2a6ad99`.

0.11.21.7 21.7 — Living docs updated

- `ROADMAP.md`: preamble updated; "Add sign correction to LB" moved to Completed; "Diagnose β□0 collapse" narrowed to YFB merged only; alpha CV stale numbers corrected.
- `DECISIONS.md`: 2026-05-05 entries for Phase C (LB) and Phase B+C (YFB).
- `README.md`: Project Status section updated to reflect both model variants and Phase C fix.

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1 Session 23 — Cohort Extension & YFB →0 Exhaustive Diagnostic (2026-05-22)

Branch: `cohort-cols-L`. All tests: 230/230 passing.

1.1 23.1 — Cohort Indicator Extension (both models)

Added a `cohort_id` parameter to both `fit_supervised_mf_modular()` (LB) and `fit_cox_on_yf()` (YFB). When supplied, C−1 binary indicator columns (corner-point encoding, reference = first alphabetical cohort) are appended to L. The corresponding F rows (`F_cohort`) are estimated via a closed-form Normal conjugate update (`update_F_cohort.R`, σ_F_cohort prior). The survival linear predictor uses only the K biological columns (β_cohort = 0 by construction).

New code: - `code/update_F_cohort.R` — `update_F_cohort_k()` and `update_F_cohort_all()` - `code/compute_elbo.R` — added `compute_normal_kl(EF, EF2, sigma_F_cohort)` for cohort prior KL - `code/fit_modular.R` and `code/fit_cox_on_yf.R` — `cohort_id` and `sigma_F_cohort` parameters - Tests: `tests/test_update_F_cohort.R` (10 tests), `tests/test_fit_modular_cohort.R` (11 tests), `tests/test_fit_yf_cohort.R` (9 tests); total 229/229 □ 230/230 with later additions

Synthetic validation (`results/cohort_lmm_sim/run_synthetic.R`): - Offset absorption: |cor(estimated offset, true offset)| = 0.995 - Factor recovery: +88–109% improvement in correlation vs. baseline (no cohort_id) - Survival concordance: correctly preserved at the biological-factor level

Real-data benchmark (`results/benchmark_sim/run_cohort_lmm_benchmark.R`): 4-way comparison: LB_base vs LB_cohort vs YFB_base vs YFB_cohort on merged TCGA+CPTAC training, evaluated on 5 external cohorts. Results in `results/benchmark_sim/outputs/cohort_lmm_benchmark/`.

Model	Mean external C (5 cohorts)	Notes
LB_base	0.618	K_eff=3
LB_cohort	0.604	K_eff=2; slightly lower on average
YFB_base	0.543	K_eff=0, β□0 on merged
YFB_cohort	0.533	K_eff=0, β□0 unchanged

LB cohort result: Gains on RNA-seq external cohorts (PACA_AU_seq +0.019, PACA_AU_array +0.007), losses on Dijk/Puleo (−0.042, −0.044). The cohort indicator absorbs the platform offset but also consumes a biological factor, reducing K_eff from 3 to 2. The K=20 regime already has ARD pruning to handle platform effects implicitly; cohort columns may be more valuable at lower K.

YFB result: Cohort extension does not resolve YFB β□0 — platform mixing drives this at the F-update level, not the L-update level.

Full evaluation report: `docs/reports/cohort_lmm_benchmark_report.{qmd,pdf,html}`.

1.2 23.2 — YFB →0 on Merged Data: Exhaustive Diagnostic

`results/benchmark_sim/run_yfb_beta_fix_diagnostic.R` tested 9 strategies (V0–V8) on merged TCGA+CPTAC (n=273, K=4 via 1-SE K-CV, max_iter=150). Two genuine bugs fixed in `fit_cox_on_yf.R` along the way; all fix strategies ultimately fail on merged data.

Bug fix 1 — SVD initialization (`pmax` □ `abs`): `pmax(svd_init$v %*% diag(d_k, K, K), 0)` zeros entire EF columns when the SVD vector points all-negative. On merged K=2 data, EF[:,1] norm was identically 0, making ZF[:,1]=0 and β_1=0 trivially. Fixed to `abs()` (equally valid for non-negative `point_exponential` prior). After fix: K-CV becomes informative (K=4 selected, mean C=0.593 vs uninformative 0.52–0.56 before).

Bug fix 2 — Cox warm-start normalization: Warm-start used Y %*% EF (un-normalized), giving β_ws ≈ 5.99e-9 at CAVI scale (machine epsilon). Fixed to normalize EF before Cox regression. After fix: β_ws ≈ [−4.5e-4, 1.3e-4].

alpha_F parameter: Wired the pre-existing `update_F_surv_YFB.R` dual-source F update through an `alpha_F` parameter (default 0, backward compatible). Previously the call was hardcoded to `alpha=0`.

Diagnostic results (V0–V8):

Strategy	K_eff	C_dijk	Notes
V0: Baseline	0	0.526	β□0
V1: Cox warm-start	0	0.521	β decays despite non-zero init
V2: β burn-in (N=10)	0	0.526	no help
V3: Warm-start + burn-in	0	0.521	no help
V4: High α=0.75	0	0.505	no help
V5: High α=0.75 + warm-start	0	0.521	no help
V6: Dual-F α_F=0.1	0	0.502	RMSE□751 (EF instability)
V7: Dual-F α_F=0.3	0	0.504	RMSE□783 (EF instability)
V8: Dual-F α_F=0.5	0	0.505	RMSE□790 (EF instability)

Root cause of V6–V8 failure (chicken-and-egg trap): With EBeta≈0, the survival gradient contribution to A_F/B_F in `update_F_surv_YFB_k()` is proportional to $E[\beta_k^2] \approx 0$, so enabling alpha_F simply adds EBNM noise to the F update without injecting survival signal. EF grows freely (instability), but β stays zero.

Conclusion: All CAVI-local fixes are exhausted on merged TCGA+CPTAC for YFB. The structural trap is β=0 □ B_beta=0: no CAVI iteration can exit unless EF already encodes survival-relevant directions, which requires β≠0 to create.

Practical landscape as of 2026-05-22:

Approach	C_dijk	K_eff
LB_merged	0.590	3
YFB_perplatform (Phase 1 preprocessing)	0.573	3
YFB_merged (any CAVI-local fix)	~0.52	0

Remaining candidate (not yet implemented): Frozen-F β pre-conditioning — freeze EF for N_frozen iterations, let β grow freely against SVD-initialized factors (breaking the β=0 fixed point), then unfreeze F. Avoids instability because EF doesn't move during β initialization.

1.3 23.3 — Open Items Entering Next Session

1. **Frozen-F β pre-conditioning** `[Priority: Medium]` — implement `N_frozen` parameter in `fit_cox_on_yf.R`: freeze EF for first N_frozen iters, update only β and L, then unfreeze. This is the one structural fix that could break the CAVI fixed-point without causing EF instability. Test on merged TCGA+CPTAC with `run_yfb_beta_fix_diagnostic.R`.
2. **Low-K cohort extension test** `[Priority: Medium]` — run `run_cohort_lmm_benchmark.R` with K_LB=5 (not K=20). At lower K, ARD cannot spare factors for platform offset absorption, so the cohort indicator may add more value. `--quick` flag in the benchmark runner uses K=5.
3. **Merge cohort-cols-L to main** — branch is in a clean state (230/230 tests, full benchmark evaluated). Decision pending on whether low-K test should run first.

4. **Re-run YFB benchmark on main** — main-branch `run_YFB_benchmark.R` used `alpha_F=0` (hardcoded). The `alpha_F` parameter is now available; consider whether any single-cohort YFB results should be updated with a small alpha_F value for stability insurance.

1.4 23.4 — Files Changed This Session

File	Change
`code/fit_cox_on_yf.R`	abs() SVD init fix; warm-start normalization fix; alpha_F param; verbose header fix
`code/update_F_cohort.R`	New — Normal conjugate cohort F update
`code/compute_elbo.R`	Added compute_normal_kl() for cohort prior KL
`code/fit_modular.R`	cohort_id / sigma_F_cohort params; Normal KL in ELBO
`tests/test_update_F_cohort.R`	10 new tests
`tests/test_fit_modular_cohort.R`	11 new integration tests
`tests/test_fit_yf_cohort.R`	9 new integration tests
`results/cohort_lmm_sim/run_synthetic.R`	Stage 1 synthetic validation
`results/benchmark_sim/run_cohort_benchmark.R`	Stage 2 real-data 4-way benchmark
`results/benchmark_sim/run_yfb_beta0_diagnostic.R`	YFB β̂0 exhaustive diagnostic (V0–V8)
`results/benchmark_sim/outputs/cohort_lmm_benchmark/`	Benchmark CSVs and RDS
`results/benchmark_sim/outputs/yfb_beta0_diagnostic/`	Diagnostic CSVs
`docs/reports/cohort_lmm_benchmark_report.{qmd,pdf,html}`	Comprehensive cohort extension report
`DECISIONS.md`	Three new entries (cohort extension, YFB β̂0 diagnosis, dual-source F)
`ROADMAP.md`	Open items updated; YFB β̂0 CAVI-local exhausted; frozen-F noted

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2 Session 22 — Four-Phase Improvement Evaluation (2026-05-06)

Branch: `pre-merge-synthesis-05_06_26` (base: `ac5b215`, "Rewrite ROADMAP and README for biostatistician readability"). Synthesis report: `docs/reports/ssbmf_summary_report_05_06_26.pdf`. All tests: 192/192 passing after adding KCV-T13 and KCV-T14.

2.1 22.1 — Phase 1: Per-Platform Standardization for YFB Merged Training

Problem: YFB merged training (TCGA_PAAD + CPTAC, K=20) collapses β̂0 (machine-epsilon). Root cause: the per-subject rank transform before quantile normalization removes the between-platform variance that survival signal depends on; A_surv/A_gen ~10^{-3} at iteration 1.

Fix: `per_platform_standardize=TRUE` + `rank_transform=FALSE` in `preprocess_desurv_cohort()`, plus K=3 (registered as `k_pdac_yfb_merged: 3` in `config/globals.yml`). Run via `--per-platform-norm --no-rank` flags in `run_YFB_benchmark.R`.

Results (K=3, both priors identical — factors escape spike at K=3):

Cohort	External C-index
Dijk (RNA-seq)	0.597
Moffitt (array)	0.532
PACA-AU (array)	0.666
PACA-AU (seq)	0.666
Puleo (array)	0.644

Median 0.644, all above 0.5, matching or exceeding DeSurv range (0.60–0.65). Results saved to `results/benchmark_sim/outputs/YFB_benchmark_perplatform/`.

2.2 22.2 — Phase 2: Initialization Constraints (Verdict: Discard)

Two variants tested on LB TCGA_PAAD (K=5, point_normal): 1. Non-negative L init: `pmax(SVD_L, 0)` — clamps negative entries to zero. 2. Cox-sign alignment: flip L columns to align with naive Cox direction.

Neither improved ELBO, K_eff, or external C-index vs. unconstrained SVD baseline. The Phase C post-fit sign correction already handles the main sign failure mode. No code retained. Verdict: **Discard**.

2.3 22.3 — Phase 3: Multi-Start Random Restarts (Verdict: Discard)

30-restart sweep on LB TCGA_PAAD (K=10, point_normal). Restart 1 = SVD initialization; restarts 2–30 = random normal, seeds 43–72.

- SVD restart (restart 1) achieves the highest ELBO in all 30 trials.
- ELBO range across restarts: 2,699 units (~0.3% relative) — nearly unimodal landscape.
- Multi-start adds ~30× computational cost with no predictive benefit.

Verdict: **Discard**. `n_init=1` (SVD) is the canonical benchmark default. Wrapper `code/fit_modular_multistar` retained as a diagnostic utility only. DECISIONS.md entry: 2026-05-06 Phase 3.

2.4 22.4 — Phase 4: K Selection via Cross-Validated C-Index

`select_K_cv()` extended with `model="LB"` (default) or `model="YFB"` parameter. LB dispatches to `fit_supervised_mf_modular` + `predict_supervised_mf`; YFB dispatches to `fit_cox_on_yf` + `predict_cox_on_yf` (passes `EF_norms`). Sign correction disabled inside CV folds for both models (fold-to-fold consistency).

5-fold CV over K□{2,…,10,15,20} on TCGA_PAAD (n=144, events=75):

Model	Prior	K_opt	Notes
LB	point_normal	8	Spike-small-n artefact (C=0.5 at K≤3)
LB	normal	3	Flat plateau (range=0.032 < SE); parsimonious
YFB	point_normal	2	β□0 collapse in all folds (C=0.5 everywhere)
YFB	normal	9	C<0.5 across all K (sign instability in CV)

Key finding: YFB normal-prior CV yields anti-concordant C-indices (0.40–0.46); mapping to 1−C gives 0.54–0.61, comparable to LB. Sign instability in YFB CV folds (without global sign correction) makes K-selection unreliable. K-CV for YFB requires fold-level sign correction before results are actionable.

Practical recommendation: K=5 for LB. K=3 is the CV-correct choice under normal prior; K=5 sits above the spike escape threshold and is more robust to prior choice.

New CSV outputs in `results/benchmark_sim/outputs/K_cv/`: `K_cv_comparison_LB.csv`, `K_cv_comparison_YFB.csv`, `K_cv_comparison_h2h.csv`, per-model-prior fold tables. `K_cv_meta.csv` updated to 4 rows (LB+YFB × PN+NM).

Tests: added KCV-T13 (YFB structure) and KCV-T14 (YFB fold count) □ 192/192 passing.

2.5 22.5 — Synthesis Report

`docs/reports/ssbmf_summary_report_05_06_26.qmd` covers all four phases. Phase 4 section includes: LB K-CV table+plot, YFB K-CV table+plot, LB vs. YFB head-to-head table+plot, and interpretation of the anti-concordance finding. Rendered to PDF and HTML. Committed: `88815b7`.

2.6 22.6 — Living Docs Updated

- `ROADMAP.md`: header updated to 2026-05-06; YFB β□0 item marked complete (Phase 1); K-CV item marked complete (Phase 4); new open item added for YFB sign correction in CV.
- `DECISIONS.md`: Phase 1 and Phase 2 entries added (2026-05-06).
- `PROJECT_STATUS.qmd`: this section (Session 22).
- `CLAUDE.md` memory: no change needed (MEMORY.md will be updated separately).

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3 Sessions 24–25 — 18-Config Benchmark, DeSurv Gene Selection, and Factor Diagnostics (2026-05-25 to 2026-05-28)

Branch: `desurv-gene-selection` (forked from `main` 2026-05-25). Tests: 246/246 passing.

3.1 24.1 — 18-Configuration Preprocessing Benchmark (2026-05-25)

Extended the 6-configuration benchmark to 18 by evaluating all combinations of preprocessing strategy × model × cohort indicator on merged TCGA_PAAD + CPTAC ($n = 273$). Five preprocessing strategies: joint QN+rank (prior standard), per-platform z-std, joint QN without rank transform, joint z-std, and log-only. CV K-selection (biological floor K≥3) applied separately for each of the 10 unique preprocessing×model combinations.

Critical finding: Per-platform z-standardization is the only viable preprocessing for pooled TCGA + CPTAC training. 10 of 12 non-per-platform configurations produce $\widehat{\beta} = 0$ for all factors (K_eff=0). The root cause is platform-dominated SVD: without per-platform z-std before merging, the top singular vectors of the merged Y matrix capture the between-cohort (TCGA vs. CPTAC) contrast, not biological variation.

Winner	Model	Preprocessing	Cohort	K	K_eff	Mean ext. C
M5	YFB	Per-platform z-std	No	3	2	0.624
M4	LB	Per-platform z-std	Yes	3	1	0.618

Files: `results/benchmark_sim/run_merged_kcv.R`, `results/benchmark_sim/run_merged_benchmark.R`, `docs/reports/merged_benchmark_report_5_25_26.{qmd,pdf}`, `docs/progress_report/SSBMF_Statu

3.2 24.2 — DeSurv Gene Selection Alignment (2026-05-27)

Implemented DeSurv-aligned gene selection in `code/preprocess_desurv.R`: - New method: `combined_rank` in `select_top_variable_genes()` — ranks genes by mean and by variance independently, retains those with lowest rank-sum (i.e., both highly expressed and highly variable). - New parameter: `selection_per_cohort=TRUE` in `preprocess_merged_cohorts()` — applies selection within each platform separately on log-transformed data before per-platform z-standardization, then intersects across cohorts. Result: 2064 genes (vs 2000 for variance-only).

Five configurations evaluated (D1–D5): D1/D2 = originals (M4/M5); D3/D4 = DeSurv-aligned LB/YFB; D5 = DeSurv-aligned YFB + cohort indicator.

Config	Model	Preprocessing	Cohort	K	K_eff	Mean ext. C
D1 (M4)	LB	Orig (var-2000)	Yes	3	1	0.616
D2 (M5)	YFB	Orig (var-2000)	No	3	2	0.624
D3	LB	DeSurv-aligned	Yes	7	2	0.622
D4 □	YFB	DeSurv-aligned	No	7	2	0.636

Config	Model	Preprocessing	Cohort	K	K_eff	Mean ext. C
D5	YFB	DeSurv-aligned	Yes	8	2	0.614

Decision: D4 adopted as new primary configuration ($\Delta C = +0.012$ vs M5, exceeds ±0.005 threshold, improves in 4/5 external cohorts). D5 cohort indicator reduces performance (–0.022 vs D4): per-platform z-std already absorbs the platform offset; the additional cohort column absorbs survival-relevant factor capacity. Decision rule applied formally — see DECISIONS.md 2026-05-27.

Files: `code/preprocess_desurv.R`, `config/globals.yml` (k_merged_lb_desurv: 7, k_merged_yfb_desurv: 7, k_merged_yfb_desurv_cohort: 8), `results/benchmark_sim/run_desurv_comparis` `results/benchmark_sim/outputs/desurv_comparison/desurv_comparison_results.csv`, `docs/reports/desurv_alignment_report_05_27_26.{qmd,pdf}`.

3.3 24.3 — Factor Diagnostics Report (2026-05-27)

Comprehensive diagnostic evaluation of D1–D5 on merged TCGA+CPTAC training cohort.

Key findings across all configurations:

- ARD shrinkage consistently identifies K_eff=2 active programs regardless of K requested (3–8), confirming the survival signal is genuinely low-dimensional in the training data.
- All five configurations identify one adverse and one protective program. This cross-formulation consistency (LB and YFB, original and DeSurv gene sets) argues strongly that the two programs reflect real biological structure, not modeling artifacts.
- D4 active factors: Factor 3 (adverse, $\hat{\beta}_3 = +0.011$, log-rank p<0.05 on training cohort) and Factor 7 (protective, $\hat{\beta}_7 = -0.041$). K-M stratification at the median loading is directionally consistent with $\hat{\beta}$ signs and replicates in all 5 external cohorts (C ≥ 0.55 in all cases).
- Schoenfeld residual test on D4 active factors: global PH assumption not violated at 0.05 level on training cohort.
- YFB $\hat{\beta}$ magnitudes are ~10–40× smaller than LB because the YFB formulation normalizes F columns to unit norm, compressing the coefficient scale; the survival signal encoded is equivalent.
- YFB models (D2, D4, D5) converge in <50 CAVI iterations; LB models (D1, D3) require up to 150 iterations.

Top-gene weight tables for D4 Factors 3 and 7 are the starting point for pathway enrichment analysis and PDAC subtype comparison (Moffitt basal/classical, Bailey et al. 2016).

Files: `docs/reports/desurv_factor_diagnostics_05_27_26.{qmd,pdf}`.

3.4 24.4 — Open Items Entering Next Session

1. **Pathway enrichment on D4 Factors 3 and 7** `[Priority: High]` — submit top-weighted genes to MSigDB hallmark / KEGG; compare patient loadings $\widehat{L}_{i,3}$ and $\widehat{L}_{i,7}$ to Moffitt basal/classical scores in TCGA_PAAD.
2. **Merge `desurv-gene-selection` to `main`** — branch ready (246/246 tests, all reports rendered and committed). Pending user confirmation.
3. **Merge `cohort-cols-L` to `main`** — older branch, evaluate vs D4 integration.
4. **YFB point_normal K-CV** still returns C=0.5 for all K — open item, not blocking.

3.5 24.5 — Files Changed

File	Change
`code/preprocess_desurv.R`	combined_rank method; selection_per_cohort parameter
`config/globals.yml`	k_merged_{lb,yfb}_desurv: 7; k_merged_yfb_desurv_cohort: 8
`results/benchmark_sim/run_desurv_comparison.R`	New runner: D1–D5 K-CV + benchmark
`results/benchmark_sim/outputs/desurv_results_*.rds`	Comparison results
`docs/reports/desurv_alignment_report_05_27_26.{qmd,pdf}`	D1–D5 external validation
`docs/reports/desurv_factor_diagnostics_05_27_26.{qmd,pdf}`	Cross-cohort programs
`docs/progress_report/SSBMF_Status_Meeting_05_28_26.{qmd,pdf}`	Updated
`tests/run_tests.R`	246/246 (2 new DeSurv alignment tests added)
`DECISIONS.md`	2026-05-27 entry: D4 adoption, D5 cohort indicator finding
`ROADMAP.md`	Header updated to 2026-05-28; diagnostics report + pathway enrichment added
`CLAUDE.md`	Recommended configuration updated to D4

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4 Session 26 — Multi-cohort simulation & 6/18 lab-meeting deck (2026-06-14 to 06-16)

4.1 26.1 — Multi-cohort simulation study (2026-06-14)

Completed the shared vs. study-specific factor-recovery simulation: EBMF-grounded ground truth, 3 scenarios (all-shared; shared & cohort-specific [hybrid]; all cohort-specific [negative control]) × 5 model arms (YFB/LB ± cohort indicator + unsupervised EBMF) × 5 seeds. The projection model recovers shared factors at |r|≈0.90–0.92 and types shared vs. study-specific with accuracy up to 1.00; in the hybrid regime it reaches C=0.81 vs EBMF□Cox 0.72; the negative control sits at chance for every arm (no fabricated signal). Proposal/results: `docs/reports/multicohort_sim_proposal_06_14_26.{qmd,pdf}`.

4.2 26.2 — Real-data unsupervised baseline (EBMF→Cox)

Added `results/benchmark_sim/run_ebmf_cox_external.R`: fits `flashier` EBMF on the merged TCGA+CPTAC training set (same gene set), then Cox on the factor scores, and scores the 5 held-out cohorts by projection. Mean external **C=0.564 (K=20)** vs YFB 0.636 — the real-data analog of the synthetic result, with far greater parsimony (K_eff=2 vs 20 factors). See DECISIONS.md 2026-06-15.

4.3 26.3 — 6/18/2026 lab-meeting deck (revised, merged to main 905279b)

Built and revised a 28-slide Quarto reveal.js deck (`presentation/walther_lab_meeting_06_18_2026/`). Per-slide PDF (`_slides.pdf`) built with **decktape** (Chrome `--print-to-pdf` was non-deterministic for reveal pagination). Three reproducible figure generators in `figs/`. A reviewer subagent passed the deck (no critical issues). Key corrections made during revision:

- **Priors restated:** point-exponential on L,F; normal on β; τ MLE (earlier drafts wrongly said "point-normal").
- **Convergence:** relative full-ELBO change < 1e-5 is the criterion (ΔL/Δβ are diagnostics) — see the clarification note in §Code Versions and DECISIONS.md 2026-06-16.
- **K vs K_eff** made explicit: K=7 retained by CV+1-SE (floor K≥3); K_eff=2 survival-active by |β|>1e-3.
- **Adverse/protective direction** read from the (YF)-projection KM (marginal), not the joint-β sign: **Program 7 adverse** (MET/ITGA3/glycolytic), **Program 3 protective** (epithelial) — opposite the β sign (suppression among correlated programs). DECISIONS.md 2026-06-16.
- Added a Limitations section and a DeSurv-comparison backup slide (grounded in Young et al., PNAS 2026).

4.4 26.4 — Files changed

File	Change
`presentation/walther_lab_meeting_06_18_2026/`	New revised deck, figures, 3 figure generators, revision notes
`results/benchmark_sim/run_ebmf_cox_external.R`	New unsupervised EBMF→Cox external baseline + outputs
`docs/plans/archive/rashidlab_meeting_feedback_04_09_26.md`	Archived
`DECISIONS.md`	2026-06-15 (EBMF baseline) and 2026-06-16 (deck corrections) entries
`ROADMAP.md`	Status → 2026-06-16; adverse/protective relabel; EBMF baseline + deck milestones

5 Session 27 — Post-lab-meeting Phase 1: objective normalization, retirement, preprocessing fix (2026-07-12)

Branch `objective-normalization` (cut from `main` at `07a429a`), merged to `main`. Implements Phase 1 of `docs/plans/archive/ssbmf_post_lab_meeting_action_plan_07_08_2026.md`, the plan that triages the Rashid lab's 6/18/2026 feedback.

5.1 27.1 — Objective normalization (Phase 1a): two rejected designs before a stable one

The genomics ELBO term is O(n·p) in aggregate, survival is O(n) — an unnormalized ~p-fold scale gap that makes `alpha`'s `(1-alpha)/alpha` mixing uninterpretable (alpha=0.5 never meant "balanced"). Two direct-rescaling attempts were tried and **empirically rejected** for the LB (Cluster A) model, at both toy (n=150,p=200) and realistic PDAC scale (n=270,p=2000):

1. **Divide genomics by p** (matches the plan's literal wording): L, F, and β collapsed to *exactly zero*, reproducibly. Cause: L and F co-adapt bilinearly (L's precision depends on F^2, F's on L^2); reducing genomics' precision removes the accidental numerical safety margin the unnormalized model relied on, and one factor's dip compounds into the other's.
2. **Multiply survival by p instead** (avoids shrinking genomics): avoided the collapse in one scenario but caused *unbounded divergence* (max|EL| > 600,000) in another — the exact fixture `tests/test_elbo.R` already used. A scheduled/ramped introduction (reusing the existing `alpha_schedule` mechanism) did not stabilize it either.

Shipped design: `genomics_divisor/survival_divisor` (new parameters on `update_L_k`, `update_F_k`, `update_beta_k`, default 1 = no-op) are never passed to `update_L_k`/`update_F_k` in either model's CAVI loop — L/F precision stays at its original scale unconditionally. Only the ELBO monitor and the beta update (no bilinear coupling) receive the rebalancing, via a new `norm_convention` parameter (`"per_p"` default = boost survival by p; `"np_n"` = the literal convention, retained for comparison). **Consequence:** LB's own alpha-interpretability goal is only partially met (documented, deferred — not a silent gap). **YFB (the production/D4 model) is unaffected by this limitation**: its L is pure-genomics-only and F defaults to `alpha_F=0`, so there is no genomics/survival competition to rebalance in the first place; it gets the full, verified-stable treatment on β and the ELBO monitor. Full investigation in DECISIONS.md 2026-07-12.

5.2 27.2 — Retire `lambda` (Phase 1b)

Removed the `lambda` survival-scale multiplier entirely (`update_L.R`, `fit_modular.R`, `fit_cox_on_yf.R` — already dead code there, `config/globals.yml`, 7 benchmark runners). `alpha` already plays the mixing-weight role `lambda` duplicated; DeSurv's own λ is an unrelated elastic-net penalty on β, a role already filled by this model's EBNM priors. Pure simplification, zero behavior change (no test referenced `lambda`; it always defaulted to a no-op 1.0).

5.3 27.3 — Fix external-cohort preprocessing mismatch (Phase 1c)

`preprocess_desurv_cohort()` unconditionally rank-transformed and never per-platform z-standardized; the DeSurv-aligned training pipeline (`preprocess_merged_cohorts`) does the opposite. Added `rank_transform`/`per_platform_standardize` parameters (defaults preserve prior behavior). Fixed at 3 call sites that needed it: `run_desurv_comparison.R` (the D4 benchmark), `run_ebmf_cox_external.R` (the EBMF 2-step baseline Phase 2 depends on — an independent-review catch, not in the original scope), and `run_YFB_benchmark.R` (CLI-flag-controlled). Checked and left alone: `run_LB_benchmark.R` (already consistent by coincidence of defaults) and `run_merged_benchmark.R` (18 heterogeneous configs reusing one external preprocessing call — a real restructuring needed, not a parameter flip; documented as a known, deferred limitation, not silently unfixed).

5.4 27.4 — Real-data verification result

Re-ran `run_desurv_comparison.R` (full, not `--quick`) on real PDAC data after all three sub-phases. **D4 external mean C-index: 0.636 → 0.642 (+0.006)**, K=7. **K_eff rose 2 → 4**: the two original active factors remain dominant and consistent in sign/magnitude with the pre-Phase-1 result (β≈+0.019, −0.042 vs. previously +0.011, −0.041), while the two newly-active factors are only marginally above `beta_threshold=0.001` (β≈+0.0034, +0.0036) — plausibly a threshold-scale artifact of Phase 1a's boosted β precision rather than a genuine parsimony loss, but not yet confirmed either way. Flagged as an open item (ROADMAP.md, CLAUDE.md) to resolve before or alongside Phase 3's own K/K_eff analysis. 264/264 core tests + 81/81 real-data tests passing throughout.

5.5 27.5 — Files changed

File	Change
`code/update_L.R`, `code/update_F.R`, `code/update_beta.R`	`genomics_divisor`/`survival_divisor` params; λ removed
`code/fit_modular.R`, `code/fit_cox_on_yf.R`	`norm_convention` param, ELBO assembly rebalanced, λ removed
`code/preprocess_desurv.R`	`rank_transform`/`per_platform_standardize` params on `preprocess_desurv_cohort()`
`config/globals.yml`	`lambda` keys removed
8 files in `results/benchmark_sim/`	λ call-site cleanup; 3 external-preprocessing fixes
`tests/test_update_L.R`, `test_update_F.R`, `test_update_beta.R`, `test_normalization.R` (new), `test_preprocess_desurv.R`, `test_fit_modular_cohort.R` (regenerated baseline)	18 new/updated tests

File	Change
DECISIONS.md	4 entries, 2026-07-12 (normalization, λ retirement, preprocessing fix)
ROADMAP.md, CLAUDE.md	Phase 1 milestone; updated D4 metrics; K_eff/beta_threshold open item

5.6 27.6 — Post-merge correction: beta boost was unjustified; fresh K-CV confirms K=7 is genuine

After merging §27.1–27.5 to main, closer review of the K_eff 2→4 open item (flagged in §27.4) found the actual root cause and corrected it, and a fresh K-CV run answered the "is K=7 really necessary" question directly rather than by inference.

Beta boost correction. Beta's own coordinate update has no genomics term competing with it in its own formula in either model (100% survival) — the genomics/survival imbalance Phase 1a targets does not structurally exist for beta. The merged design nonetheless boosted beta's precision by p (justified at the time as "safe," which is not the same as "necessary"). Confirmed empirically on real data: with the corrected default boost_beta=FALSE, D4's β values are essentially identical to the pre-Phase-1 baseline, K_eff reverts to 2, and the honest external C-index is **0.627** (not the previously-reported 0.642 — that number was an artifact of the unjustified boost). The 0.636→0.627 movement is attributable entirely to Phase 1c's preprocessing fix, not to Phase 1a, which now has zero effect on YFB's fitted output (ELBO-monitor-only). **Honest conclusion: Phase 1a's objective normalization provides no performance benefit for YFB** — there was no real per-coordinate imbalance there to begin with. It's kept anyway as a genuine, if purely internal, correctness fix to the ELBO monitor used for convergence.

Fresh K-CV (YFB, D4 preprocessing, K=2:10, no floor). K=7 is genuine: a real, non-noise jump exists between K=6 (mean C=0.593) and K=7 (0.633), larger than the ~0.03 fold-to-fold SE; K=8 is the actual peak (0.651); K=2–6 are all meaningfully worse, not just noisier. Kept K=7. The 7-vs-DeSurv's-3 gap traces to two confirmed methodology differences, not evidence the data itself needs more capacity: DeSurv jointly tunes k/α/λ via Bayesian optimization with a fixed elastic-net penalty, while we tune only K via CV with α fixed and no penalty. K_eff=2 already tracks DeSurv's own ~1 survival-active factor reasonably well — the parsimony story that holds up is about *effective*, not total, factors. A fair K comparison needs a matched-protocol DeSurv re-run (Phase 2/6), not a different K-CV on our side.

Full reasoning, the empirical comparison table, and the CV table are in DECISIONS.md 2026-07-12 (two new entries). CLAUDE.md, ROADMAP.md corrected to match.

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6 Session 28 — Phase 2: joint model vs. two-step value-add (2026-07-12)

Branch `validation-two-step` (cut from `main` after Phase 1's merge and post-merge correction). Implements Phase 2 of `docs/plans/archive/ssbmf_post_lab_meeting_action_plan_07_08_2026.md`.

6.1 28.1 — PCA+Cox two-step baseline

New, TDD-developed `results/multi_cohort_sim/fit_pca_cox.R`: unsupervised PCA on training genomics data, then Cox regression on the top-K PC scores; held-out scoring reuses the training rotation and center. 9/9 tests (structure, train/test projection consistency, degenerate-fit handling, signal recovery on a known prognostic direction).

6.2 28.2 — Survival-strength sweep

New `results/multi_cohort_sim/run_survival_strength_sweep.R` and a `synthetic_multicohort.su` config block: scales the true prognostic effect (`beta_shared`) from 0 to large across 6 levels × 5 seeds, holding the genomics DGP fixed. Compares YFB (joint, tuned α=0.5) against PCA+Cox and EBMF+Cox (existing) two-step baselines, plus LB (joint) as a secondary reference and YFB at α=0 as the internal control.

Honest results (full report: `docs/reports/joint_vs_twostep_sweep_07_12_2026.{qmd,pdf,html}`): - **Equivalence at zero signal:** YFB=0.517±0.008 vs. PCA+Cox=0.533±0.017 vs. EBMF+Cox=0.540±0.017 — within ~1 SE of each other and of chance; consistent with equivalence though the raw gaps (0.016–0.023) sit slightly above an idealized <0.01 target. - **Separation growing with signal:** not immediate — tied with baselines at strength=0.25 — but reliable and growing from strength≥0.5 onward (0.931 vs. 0.908/0.875 at the strongest level, with YFB's own SE shrinking from 0.027 to 0.006 as signal strengthens). - **α=0 internal control (cleanest result):** max|F(α=0.5) − F(α=0)| = exactly 0 across all 30 strength×seed combinations. Required controlling for a real confound first: naturally-converged fits at different α values stop after different iteration counts (different combined-objective trajectories), so a naive F comparison contrasts different snapshots of an otherwise-identical trajectory, not a true α effect. Forcing identical iteration counts (`tol=-1`) isolates the genuine structural property cleanly.

6.3 28.3 — Files changed (initial pass)

File	Change
`results/multi_cohort_sim/fit_pca_cox.R` (new)	`fit_pca_cox()`/`predict_pca_cox()`
`results/multi_cohort_sim/run_survival_strength_sweep.R` (new)	The sweep runner
`config/globals.yml`	New `survival_strength_sweep` config block
`tests/test_fit_pca_cox.R` (new)	9 tests

File	Change
`docs/reports/joint_vs_twostep_sweep_report_2026.{qmd,pdf,html}` (new)	Full report
`DECISIONS.md`, `ROADMAP.md`	Phase 2 entry/milestone

6.4 28.4 — Comprehensive extension, same day: 10 seeds, 4 DGP scenarios

The initial 5-seed/1-scenario pass was directionally right but too thin to be strong evidence. Extended `run_survival_strength_sweep.R` and `config/globals.yml` (`survival_strength_sweep.{n_see` to run 10 seeds across 4 DGP scenarios: `default` (real EBMF templates, as before), `sparse_synthetic` (synthetic sparse F, 2% active genes/factor, real templates off), `low_snr` (shared-factor amplitude 12→6), `high_K` (K_shared 4→8).

`default`/`low_snr`/`high_K` confirm the original pattern (equivalence at zero, growing joint advantage from moderate signal, exact α-invariance — now 240/240 combinations, not 30/30).

`sparse_synthetic` reverses the ordering: at strength=4, YFB=0.795±0.027 vs. PCA+Cox=0.934±0.005 vs. EBMF+Cox=0.940±0.004 — roughly 5 SEs below both two-step baselines, not noise.

Root-caused with a persisted, reproducible diagnostic (`results/multi_cohort_sim/diagnose_factor_collapse.R`, 5 seeds), not left as an open question: 1. YFB at α=0 (survival term fully removed) shows the *identical* dead-factor count as the tuned fit in every one of 5 seeds (mean 1.8/4 dead, both cases) — rules out the joint survival objective as the cause. 2. LB (same `update_L.R`/`update_F.R` machinery) also collapses — on average *more* factors die than YFB (mean 2.4/4 vs. 1.8/4 over 5 seeds; worse in 3/5, better in 2/5) — confirms a shared CAVI vulnerability, not something specific to the Cox-on-YF reformulation, though severity is seed-dependent rather than uniformly worse for LB. 3. Random init makes it uniformly worse (4/4 dead in every seed vs. 1–2/4 for SVD init) — rules out "bad initialization" as the fixable cause.

Diagnosis: NOT an amplitude-hierarchy effect (`a_shared` is identical across all shared factors in every scenario, including `default` — verified directly). The actual differentiator is loading structure: real EBMF templates are dense (97% of genes non-trivially loaded) with graded magnitudes and real cross-factor correlation (up to |r|≈0.4), while `sparse_synthetic` gives each factor exactly 2% active genes, uniform magnitude, and near-zero cross-factor overlap by construction. This near-perfect symmetry appears to be exactly the condition under which our point_exponential joint CAVI hits a degenerate fixed point — a factor that dips slightly below its (otherwise identical) siblings gets shrunk further every iteration with no recovery mechanism, and the ELBO genuinely plateaus there (fast "convergence" at a bad solution). EBMF avoids this via greedy, residual-based factor-at-a-time fitting; PCA avoids it by doing no shrinkage at all.

Scoping — narrower than "only a synthetic-data artifact": `default` (real templates) showed zero collapses for either model (0/60 fits). But in `low_snr`/`high_K` — also built on real EBMF templates — **LB collapsed completely on 10 of 180 fits; YFB on 0 of 180**. So YFB has not collapsed under any real-template scenario tested here, but LB has, occasionally, under lower SNR or higher K — this is not purely a contrived-synthetic-data phenomenon, though there is still no evidence it affects the real PDAC fits. Logged as a genuine, understood algorithmic limitation (ROADMAP.md follow-up item, priority elevated accordingly), not resolved here.

Additional files changed:

File	Change
`results/multi_cohort_sim/run_survival_strength_sweep.R`	Scale strength, 10 seeds.
`results/multi_cohort_sim/plot_survival_strength_sweep.R`	Facet by scenario (script itself created in an earlier commit; modified here for faceting)
`results/multi_cohort_sim/diagnose_fold_cause_signal.R` (new)	Fold-cause diagnostic for Result 4
`config/globals.yml`	`n_seeds`, `scenarios` block
`docs/reports/joint_vs_twostep_sweep_results_2026-07-11.{qmd,pdf,html}`	Result 4 (scenario-stratified) + diagnosis
`DECISIONS.md`, `ROADMAP.md`	Same-day follow-up entry + new CAVI-collapse and Phase 3 items

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7 Session 29 — Phase 3: K-parsimony curve on real data (2026-07-13)

Branch `phase3-k-parsimony` (cut from `main`, independent of Phase 2's unmerged `validation-two-step`). The internal K-CV table (previous session) measures training-fold C-index; it doesn't say how much of the recommended config's **external** validation number (D4: mean C=0.627, K=7) would survive at a smaller, more parsimonious K.

7.1 29.1 — K-parsimony curve

New `results/benchmark_sim/run_k_parsimony_curve.R`: refits YFB D4 (per-platform z-std, DeSurv combined-rank gene selection, top-3000 per cohort, no cohort_id) at K ∈ {2, 3, 4, 5, 7}, re-running external validation against the same 5 held-out PDAC cohorts for each K.

Result:

K	Mean external C	SE	K_eff
2	0.541	0.012	1
3	0.594	0.016	1
4	0.541	0.012	1
5	0.596	0.027	3
7	**0.627**	0.020	2

Applying the same 1-SE rule used for internal K-CV: K=7's margin is 0.627−0.020=0.607, and no smaller K reaches it (K=5's 0.596 is closest, still short). **K=7 is not free to shrink, in this single-seed comparison** — this is the honest answer, not the parsimony win we might have hoped for.

Two things worth noting rather than glossing over: - **K=7 wins on every one of the 5 individual external cohorts** (not just on average) — a consistent, non-cherry-picked result. - **The curve is**

non-monotonic and K_eff doesn't track it: K=4 (K_eff=1) essentially ties K=2 (K_eff=1) despite one more factor; K=5 (K_eff=3, more "active" factors than K=7's 2) still underperforms K=7. K_eff alone is not a reliable predictor of external performance here.

Caveat (raised by independent review, not glossed over): each K used a single seed/init. K=2 and K=4 converge suspiciously fast (7 and 9 iterations) to a tiny beta_max (~0.009), while K=3/K=7 take 20 iterations to a substantially larger beta_max (0.033/0.040) and K=5 takes 41 — the fast, near-zero-β pattern at K=2/K=4 is consistent with the same CAVI factor-collapse failure mode root-caused in Session 28 (Phase 2's `sparse_synthetic` finding), not necessarily a genuine ceiling for those K values. A multistart (best-ELBO) rerun at K=2/K=4 would be needed before treating those specific numbers as conclusive. K=5's result is less suspect (converges normally) and already falls short of K=7 on its own, so the qualitative conclusion likely survives, but the precise K=2/K=4 numbers should not yet be treated as a hard ceiling.

Conclusion: the K=7-vs-DeSurv's-K=3 gap is not simply "extra factors that could be pruned for free" — refitting at smaller K directly (not just counting active factors from the K=7 fit) shows real performance loss at every smaller K tested here, on every held-out cohort, with the caveat above applying most to K=2/K=4 specifically. Whether DeSurv's joint (K,α,λ) Bayesian-optimization tuning would find a smaller K with comparable *external* performance remains untested — a plan for this exists (`docs/plans/archive/joint_k_alpha_bayesopt_plan_07_12_2026.md`), not yet implemented.

7.2 29.2 — Files changed

File	Change
`results/benchmark_sim/run_k_parsimony.R` (new)	K-parsimony curve runner
`results/benchmark_sim/outputs/k_parsimony_curve_results.csv` (new)	Raw results (5 K × 5 cohorts)
`DECISIONS.md`, `ROADMAP.md`	Phase 3 entry/milestone

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8 Session 30 — Pathway enrichment on the recommended model's active gene programs (2026-07-15)

(Note: the K-parsimony follow-up plan's Steps 1-4, completed 2026-07-13 between this entry and Session 29 above, are logged in `DECISIONS.md` 2026-07-13 but do not yet have a corresponding session entry here — out of scope for this session's work, flagged so the gap is explicit rather than silently skipped.)

Branch `pathway-enrichment-plan` (cut fresh from `main`; the original branch of the same name predated ~8,000 lines of subsequent work and was unusable as a base — only its plan document, `docs/plans/archive/pathway_enrichment_plan.md`, was retrieved and carried forward). Goal: assign biology to Program 7 (Adverse) and Program 3 (Protective), the recommended model's two survival-active ($K_{\text{eff}} = 2$) gene expression programs.

8.1 30.1 — Enrichment engine and custom gene sets

New `code/pathway_enrichment.R`: `load_d4_weights()`, `run_fgsea_program()` (rank-based, one-sided via `scoreType="pos"` since the point-exponential **F** prior makes all weights ≥ 0), `run_ora_program()` (confirmatory, top-N genes), `build_pdac_genesets()` (Moffitt basal/classical, Bailey 4-subtype, DeSurv D1-D3 — all recovered from local reference data / the DeSurv SI appendix, not re-derived from memory), `get_msigdb_collections()` (Hallmark/Reactome/KEGG/GO:BP via `msigdbr` 26.1.0 — note this version renamed `category`/`subcategory` to `collection`/`subcollection`).

DeSurv's D1/D2/D3 gene lists (270 genes each) are parsed directly from the SI appendix PDF's text (`pdftotext -layout`) — two real parsing bugs were caught before they silently corrupted the gene lists: page-footer page numbers were being picked up as spurious "genes," and mixed-case/hyphenated symbols (`C19orf33`, `HLA-H`) were dropped by an overly strict regex. Both fixed; verified all 3 factors parse to exactly 270 unique genes with zero cross-factor overlap.

8.2 30.2 — Enrichment result and full production run

All 7 programs enriched (sanity check); Programs 5-6 (nominally survival-inactive) still enrich for real stromal/immune biology (12 and 40 coherent hits respectively) — genuine variance axes in the data that just aren't survival-predictive here, not noise. Program 7's headline associations: basal-like/squamous tumor identity (DeSurv D3, Bailey Squamous) and MET/EGFR-driven RTK signaling. Program 3's: classical/differentiated tumor identity (DeSurv D1, Moffitt Classical), with a secondary, weaker basal-like association reported honestly rather than smoothed over (§30.5).

F1's dot plot needed a real fix, caught visually: MSigDB's full systematic set names are long enough that an 11x6.5in figure squeezed the actual data points into an invisible sliver of the plot area. Fixed (16x8in @150dpi). F3's gene-weight heatmap had the same problem in the other direction — 113 gene-name row labels overlapping illegibly at a fixed figure size — fixed with a `heatmap_dimensions()` helper that scales font size down and figure height up with gene count.

8.3 30.3 — PDAC subtype concordance (Step 7)

Matched D4's pooled patient loadings back to the 144 TCGA-PAAD training samples (`load_pdac_raw()` previously discarded sample IDs via `rownames(Y) <- NULL`; added a `sampID` field, additive and non-breaking). Caught and corrected a real error in the original plan draft: its decision pointed to "MS"/"MS_K2" as the per-sample tumor basal/classical label, but direct inspection shows those columns are the Moffitt **stroma** activation axis (Activated/Normal), not tumor subtype. Used PurIST (the correct, already-available axis) instead. Result: 144/144 matched (100%); Program 7 correlates positively with basal-likelihood (Spearman $\rho = +0.58$,

$p = 2.1 \times 10^{-14}$), Program 3 negatively ($\rho = -0.57$, $p = 4.9 \times 10^{-14}$) — both far exceeding significance.

8.4 30.4 — External cohort robustness (Step 8)

A TDD test initially caught a real sign-convention issue before it touched real data: `survival::concordance(Surv(...) ~ score)` treats a *higher* predictor as predicting *longer* survival by default. The first fix (a fixed `I(-score)` negation, matching `code/select_alpha_cv.R`/`code/predict.R`) turned out to only be correct for an adverse signature — independent code review (§30.7) caught that it silently reports the *wrong-direction* C-index for a protective one. Final, correct fix: match this project's own `oriented_cindex()` convention (`max(c_raw, 1-c_raw)`, direction-agnostic). Result after the final fix: Program 7's signature has HR>1 in **5/5** external cohorts (range 1.30-2.40, C-index 0.54-0.60); Program 3's has HR<1 in **5/5** cohorts (range 0.42-0.88, C-index 0.51-0.61) — exceeding the plan's own bar of "a majority of cohorts," and both programs' C-index now correctly >0.5 throughout.

8.5 30.5 — SBMF-vs-DeSurv overlap and direction sanity (Steps 9-10)

Direct gene-list Jaccard/hypergeometric overlap (top-270 genes per program vs. each DeSurv factor, independent of the fgsea/ORA results, to avoid circularity) reproduces the same story: Program 7 overlaps DeSurv D3 most strongly (80/270, $p = 3.2 \times 10^{-16}$); Program 3 overlaps D1 most strongly (61/270, $p = 1.1 \times 10^{-7}$), with the same secondary D3 association also seen in §30.2 (49/270, $p = 2.8 \times 10^{-3}$) — a real, twice-independently-reproduced nuance (and, per §30.2's ORA cross-check, weaker under that method too), not noise, and not a contradiction of the dominant classical/protective signal. All five methods run (fgsea enrichment, its ORA confirmatory check, subtype concordance, external-cohort survival, gene overlap) agree with no contradictions.

8.6 30.7 — Independent code review before merge

Dispatched `superpowers:code-reviewer` over the full branch diff before merging. Found and fixed two real bugs (both re-verified end to end — re-ran the affected analyses on real data and re-rendered the report, rather than assuming the fix was safe):

1. **[Critical]** §30.4's original `cohort_signature_cox()` fix (fixed `I(-score)` negation) was correct for Program 7 (Adverse) but silently wrong-direction for Program 3 (Protective) — the originally-published C2 table showed Program 3's C-index at 0.39-0.49 in all 5 cohorts ("worse than chance") when its true discriminative accuracy in the correct direction was 0.51-0.61. Fixed to `max(c_raw, 1-c_raw)`, matching this project's existing `oriented_cindex()` convention used in 8+ other benchmark scripts. New regression test (T9.5) constructs a protective signature specifically.
2. **[Important]** `compute_geneset_overlap()`'s hypergeometric test used each gene list's unrestricted length, but DeSurv's 270-gene lists are not a full subset of the 2064-gene SBMF background (~245-259 of 270 actually fall inside it) — understating significance by up to ~22x (Program 3 vs. D1: p moved from ~2.4e-6 to the correct ~1.1e-7). Fixed by restricting both

gene lists to the background before computing overlap/union/phyper(). New regression test (T10.3b).

Also addressed: added the ORA confirmatory-check comparison to the report (it had been computed in §30.2 but never actually compared against fgsea, despite being built for exactly that purpose — review finding #3), and added a fail-loud guard to run_external_cohort_robustness.R's headline selection matching its sibling script's existing pattern (review finding #4). Neither bug changed any directional conclusion (HR signs, Spearman ρ signs, and which DeSurv factor each program overlaps most were all unaffected) — both were precision/exact-number issues, not correctness-of- conclusion issues, but both made a published table actively misleading as originally computed.

8.7 30.8 — Files changed

File	Change
code/pathway_enrichment.R (new)	All reusable enrichment/concordance/overlap functions
results/benchmark_sim/run_pathway_fgsea_ORA.R (new)	fgsea/ORA runner, all collections, F1-F3
results/benchmark_sim/run_subtype_ORA_concordance.R (new)	ORA concordance, F4/T3
results/benchmark_sim/run_external_cohort_robustness.R (new)	5-cohort survival, C2
results/benchmark_sim/run_sbmf_descriptive_overlap.R (new)	Gene-overlap, F5/T4
results/benchmark_sim/generate_pathway_enrichment_figures.R (new)	Regenerate F1-F3 from cached results
results/benchmark_sim/benchmark_helpers.R	Added sampID field to load_pdac_raw()'s return
docs/reports/pathway_enrichment_results.{qmd,pdf,html} (new)	Full analysis report
docs/plans/archive/pathway_enrichment_plan.md (new)	Retired plan document
tests/test_pathway_enrichment.R (new, 44 tests)	Main suite: 355/355 (was 311/311 at session start)
tests/test_real_data_loading.R (+11 tests)	Real-data suite: 88/88 (was 77/77)
DECISIONS.md, ROADMAP.md, CLAUDE.md	Session entry, milestone, quick-reference updates

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8.8 Chapter 4 continuing-work framing (2026-10-02)

At the author's request, the prelim chapter's dated timeline table was removed and retained in paper/prelim/notes/removed-timeline-2026-10-02.md. The continuing-work section now

describes additional methodology and expected evidence without calendar commitments: survival-informed program discovery, feature-count/screening sensitivity, multimodal rank selection and external predictive and biological evaluation. Existing methods and results are unchanged. The advisor-meeting follow-up plan will guide a later substantive chapter update after methodological decisions and additional results are verified. The installed sync hook regenerates Chapter 4 in bios-dissertation; the full prelim must then be rebuilt there.